

SIGNALS, FAST

SAMPLING · LTI SYSTEMS · THE Z-TRANSFORM · THE DFT AND THE FFT

Thirteen chapters, each one a screen or two — the night-before version of the course, not the course. Every chapter closes with five lines that are the whole chapter, and the switch in the margin hides everything except those lines.

Read it forwards once, then read only the summaries.

13	62	23	75
CHAPTERS	SECTIONS	DIAGRAMS	SUMMARY LINES

CHAPTER 01

SIGNALS, AND WHAT DISCRETE TIME CHANGES

A signal is just information written as a function of time. Sample it and two things change that catch people out every year: frequency stops being unbounded, and a sinusoid stops being automatically periodic. This chapter settles the classification of signals, the energy-versus-power test, and the one-line periodicity test the paper likes to open with.

1.1 The classes that matter

Sort a signal first. The method that works on one class often fails on another, and the exam asks you to name the class before anything else.

- **Continuous-time or discrete-time** — is the time axis a line or a list of whole numbers? Write $x(t)$ for one, $x(n)$ for the other.
- **Deterministic or random** — can you write a formula for every future value, or only a probability?
- **Periodic or aperiodic** — does it repeat exactly after some N ?
- **Energy or power** — does the total squared value settle to a finite number, or does only the average settle?
- **Even or odd** — mirror image about $n = 0$, or anti-mirror.

1.2 Energy signals and power signals

In discrete time, adding up replaces integrating. Two sums, and a signal can only be one of the two kinds.

$$E = \sum_{n=-\infty}^{\infty} |x(n)|^2$$

$$P = \lim \left(\frac{1}{2N+1} \right) \sum_{n=-N}^N |x(n)|^2$$

periodic signal : $P = \left(\frac{1}{N} \right) \sum_{n=0}^{N-1} |x(n)|^2$

THE TWO-WAY RULE

If E is finite then P = 0, and the signal is an **energy signal**. If E is infinite but P is finite and non-zero, it is a **power signal**. Anything periodic is a power signal, because it never dies away. A decaying exponential is an energy signal.

1.3 Frequency: continuous against discrete

For an analog sinusoid $\cos(2\pi Ft)$ the frequency F can be anything at all, and every different F is a different signal. Once the time axis is a list of integers none of that survives. Write the digital frequency as $f = \frac{F}{F_s}$, or in radians per sample as $\omega = 2\pi f$.

QUESTION	ANALOG $\cos(2\pi Ft)$	DIGITAL $\cos(\omega N)$
Range of frequency	$-\infty$ to $+\infty$, all distinct	Only one period is distinct: $-\pi \leq \omega \leq \pi$, that is $-\frac{1}{2} \leq f \leq \frac{1}{2}$
Adding 2π to the frequency	A genuinely different signal	The same signal — ω and $\omega + 2\pi k$ are indistinguishable
Always periodic?	Yes, period $1/F$	Only when f is a ratio of whole numbers
Fastest oscillation	No fastest — F can keep rising	At $\omega = \pi$ ($f = \frac{1}{2}$), one sign change per sample
Distinct harmonics	Infinitely many	Exactly N of them, then they repeat

1.4 The periodicity test

THE TEST, IN ONE LINE

$\cos(\omega_0 n + \phi)$ is periodic **only if** $\frac{\omega_0}{2\pi}$ is a ratio of two whole numbers. Write $f = \frac{k}{N}$ in lowest terms; then **N is the fundamental period**. For a sum of sinusoids, take the LCM of the individual periods.

THE TRAP

$\cos(3n)$ is *not* periodic. Here $f = 3/(2\pi)$, and π is irrational, so no whole number of samples ever brings the wave back to where it started. The continuous signal $\cos(3t)$ is perfectly periodic — sampling destroyed it.

WORKED EXAMPLE

Say whether each is periodic, give the period, and say whether it is an energy or a power signal.

$$(a) : x(n) = \cos\left(\frac{\pi n}{8} + 0.2\right)$$

$$f = \frac{\frac{\pi}{8}}{2\pi} = \frac{1}{16} \text{ ratio of whole numbers} \rightarrow \text{periodic}$$

$$N = 16 \text{ (the phase } 0.2 \text{ shifts it, it cannot change } N)$$

$$P = \text{average of } \cos^2 \text{ over one period} = 0.5, E = \text{infinite}$$

$$> \text{power signal, } \frac{\text{average power } 1}{2}$$

$$(b) : x(n) = \cos(3n)$$

$$f = \frac{3}{2\pi} \text{ irrational} \rightarrow \text{NOT periodic}$$

bounded but never decays $\rightarrow$ still a power signal

$$(c) : x(n) = \left(\frac{1}{2}\right)^n u(n)$$

$$E = \sum_{n=0}^{\infty} \left(\frac{1}{4}\right)^n = \frac{1}{1 - \frac{1}{4}} = \frac{4}{3} \text{ finite}$$

$$> \text{energy signal, } E = \frac{4}{3}, P = 0$$

$$(d) : x(n) = \cos\left(\frac{\pi n}{3}\right) + \sin\left(\frac{\pi n}{4}\right)$$

$$\text{first term : } f = \frac{1}{6} \rightarrow N_1 = 6$$

$$\text{second term : } f = \frac{1}{8} \rightarrow N_2 = 8$$

$$N = \text{LCM}(6, 8) = 24 \rightarrow \text{periodic, } N = 24$$

Checked numerically: $x(n)$ in (d) repeats at a lag of 24 and at no smaller lag — a lag of 12 does **not** reproduce it.

EXAM TIP

Periodicity and energy-versus-power come as one four-mark opener. Answer in this order every time: write f , say rational or not, give N or say “not periodic”, then one sentence on energy versus power. Four lines, four marks.

CHAPTER 01 AT A GLANCE

- Digital frequency $f = \frac{F}{F_s}$; only $-\frac{1}{2} \leq f \leq \frac{1}{2}$ is distinct, and ω and $\omega + 2\pi k$ are the same signal.
- A discrete sinusoid is periodic **only if** f is a ratio of whole numbers; write $f = k/N$ in lowest terms and N is the period.
- For a sum of sinusoids the period is the **LCM** of the individual periods.
- Energy signal $\Rightarrow E$ finite and $P = 0$. Power signal $\Rightarrow E$ infinite and P finite. Every periodic signal is a power signal.
- The fastest discrete oscillation is at $\omega = \pi$: one sign change per sample. There is nothing faster.

CHAPTER 02

SAMPLING, ALIASING AND QUANTIZATION

Sampling is the one step where information can be lost for good. This chapter settles how fast you must sample, exactly which frequency a too-fast signal comes back as, and what the extra error from rounding each sample costs you in decibels.

2.1 The sampling relation

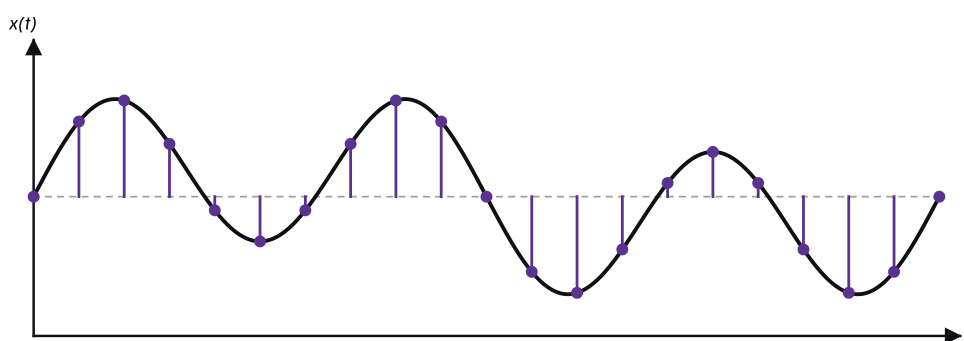
Take a reading every T_s seconds. The analog signal becomes a list of numbers, and the analog frequency F becomes the digital frequency f .

$$x(n) = x_a(nT_s) \quad T_s = \frac{1}{F_s}$$

$$f = \frac{F}{F_s} \quad \omega = 2\pi f = \Omega T_s$$

$$-\frac{1}{2} \leq f \leq \frac{1}{2} \longleftrightarrow -\frac{F_s}{2} \leq F \leq \frac{F_s}{2}$$

DIAGRAM



each stem is one sample, taken every T_s seconds

$f_s \geq 2 f_m$ — the Nyquist rate. Below it the samples no longer decide the signal.

Fig. 2.1 One analog sinusoid with sample instants marked every T_s , the stem plot of $x(n)$ drawn below it, and the two frequency axes stacked so that $F = F_s/2$ lines up with $f = 1/2$. The picture should make the substitution $f = F/F_s$ visible rather than algebraic.

2.2 The sampling theorem

THE THEOREM

A signal band-limited to $F_{\max}$ is recovered **exactly** from its samples provided $F_s > 2F_{\max}$. The number $2F_{\max}$ is the **Nyquist rate**. Half the sampling rate, $\frac{F_s}{2}$, is the **folding frequency** — the highest frequency the sampled signal can represent.

TWO WORDS THAT GET SWAPPED

The **Nyquist rate** is a property of the *signal*: twice its highest frequency. The **folding frequency** is a property of the *sampler*: half its rate. Saying “Nyquist frequency” for either one is how marks go missing.

2.3 Aliasing: which frequency you actually get

Every analog frequency in the family $F_0 + kF_s$ lands on exactly the same set of samples. If a component sits above the folding frequency it does not vanish — it comes back wearing a different, lower frequency, and nothing downstream can tell the difference.

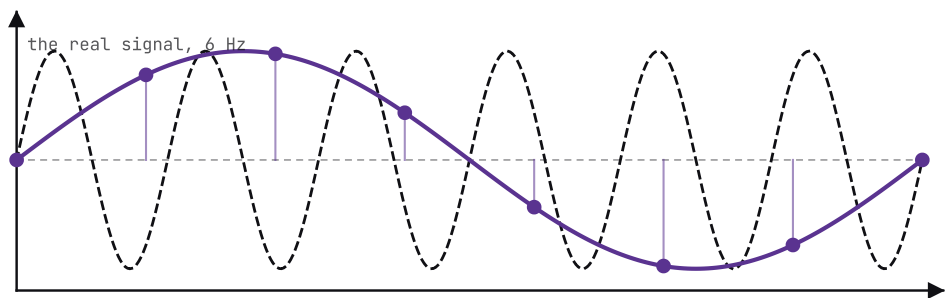
aliases of F_0 : $F_0 + kF_s$ for every integer k

to find the apparent frequency:

$f = F/F_s$, then add or subtract whole numbers until $-1/2 \leq f \leq 1/2$

apparent $F = |f| * F_s$

DIAGRAM



what the samples say, 1 Hz

both curves pass through every sample, so once the samples are taken the difference is gone for good

Fig. 2.2 A 600 Hz cosine and a 100 Hz cosine drawn on the same axes, with sample dots at $F_s = 500$ Hz sitting on *both* curves at once. The caption of the drawing is the whole lesson: from the dots alone the two are the same signal.

WORKED EXAMPLE

The signal $x_a(t) = 3 \cos(2\pi 100t) + 2 \cos(2\pi 300t) + \cos(2\pi 600t)$ is sampled at $F_s = 500$ Hz. What comes out?

folding frequency = $F_s/2 = 250$ Hz

100 Hz : $f = 100/500 = 0.20$	already inside $\pm 1/2$	$\rightarrow$ 100 Hz
300 Hz : $f = 300/500 = 0.60$	$0.60 - 1 = -0.40$	$\rightarrow$ 200 Hz
600 Hz : $f = 600/500 = 1.20$	$1.20 - 1 = +0.20$	$\rightarrow$ 100 Hz

So the sampled signal contains only 100 Hz and 200 Hz. Worse, the 600 Hz component has landed **on top of** the genuine 100 Hz one and added to it. That damage cannot be undone afterwards.

F_s needed for no aliasing = $2 F_{\max} = 2 \cdot 600 = 1200$ Hz

Verified by sampling all three cosines: $\cos(2\pi 600t)$ and $\cos(2\pi 100t)$ agree at every sample instant, and so do $\cos(2\pi 300t)$ and $\cos(2\pi 200t)$.

EXAM TIP

Always answer an aliasing question in Hz *and* say which components collide. “600 Hz appears as 100 Hz” is half a mark; “600 Hz appears as 100 Hz and is therefore inseparable from the true 100 Hz component” is the full mark.

2.4 Quantization and coding

Sampling fixed the time axis. Quantization rounds each sample to one of L allowed levels, and coding writes that level in b bits, so $L = 2^b$. This step is where the *error* comes from — and unlike aliasing it is always present.

$$\text{step size } D = \frac{x_{\max} - x_{\min}}{L - 1}$$

$$\text{error bound } |e_q(n)| \leq \frac{D}{2}$$

$$\text{SQNR(dB)} = 1.76 + 6.02b$$

THE NUMBER TO REMEMBER

Each extra bit buys about **6 dB** of signal-to-quantization-noise ratio. At $b = 8$ that is 49.9 dB; at $b = 16$ it is 98.1 dB. If a question says “how many bits for 60 dB”, solve $60 = 1.76 + 6.02b$ and round up — $b = 10$.

CHAPTER 02 AT A GLANCE

- Sample faster than twice the highest frequency: $F_s > 2F_{\max}$. The Nyquist rate is $2F_{\max}$; the folding frequency is $\frac{F_s}{2}$.
- Every frequency in the family $F_0 + kF_s$ gives identical samples — that is aliasing.
- To find the apparent frequency: form $f = F/F_s$, slide it into $-\frac{1}{2} \leq f \leq \frac{1}{2}$ by whole numbers, multiply back by F_s .
- Aliasing is **not** recoverable: two components can land on the same digital frequency and add.
- Quantization step $D = \frac{x_{\max} - x_{\min}}{L-1}$, error at most $D/2$, and SQNR = $1.76 + 6.02b$ dB — about **6 dB per bit**.

CHAPTER 03

DISCRETE-TIME SIGNALS: WRITING THEM, MOVING THEM

Short chapter, but it is the grammar for everything after it. Four ways to write one signal, the handful of elementary signals every answer is built from, and the one operation ordering — fold against shift — that wrecks more convolution answers than any arithmetic slip.

3.1 Four ways to write the same signal

- 1 **Functional** — $x(n) = (0.5)^n$ for $n \geq 0$, 0 otherwise.
- 2 **Tabular** — two rows, n on top and $x(n)$ underneath.
- 3 **Sequence** — $x(n) = \{1, 2, 3, 4\}$ with an arrow marking $n = 0$.
- 4 **Graphical** — the stem plot.

THE ARROW IS NOT DECORATION

In sequence notation the arrow says where $n = 0$ sits. Without it $\{1, 2, 3\}$ is three different signals. Track the arrow through every shift, fold and convolution — losing it is the single most expensive habit in this subject.

3.2 The elementary signals

unit impulse $d(n) = 1$ at $n = 0$, else 0
 unit step $u(n) = 1$ for $n \geq 0$, else 0
 unit ramp $r(n) = n$ for $n \geq 0$, else 0
 exponential $x(n) = a^n u(n)$

$$\text{links : } d(n) = u(n) - u(n-1) \quad u(n) = \sum_{k=-\infty}^n d(k)$$

THE THREE SIFTING FACTS

$x(n)d(n-k) = x(k)d(n-k) \cdot \sum x(n)d(n-k) = x(k) \cdot$
 $x(n) = \sum_k x(k)d(n-k)$. The third one is the whole reason convolution exists: it writes any signal as a row of weighted impulses.

3.3 Operations on the index

Three things can be done to n : shift it, fold it, scale it. Shifting by k gives $x(n-k)$ — a **delay** to the right for positive k . Folding gives $x(-n)$ — a mirror about $n = 0$. Scaling by an integer M gives $x(Mn)$, which throws samples away, and $x(\frac{n}{M})$, which needs samples invented in between.

FOLD AND SHIFT DO NOT COMMUTE

$x(-n - k)$ is **not** $x(-n + k)$. Always rewrite the argument in the form $x(-(n + k))$ or $x(-(n - k))$ first, then fold, then shift by whatever is left inside the bracket. Doing it by eye gets the answer reflected to the wrong side of the origin.

WORKED EXAMPLE

Take $x(n) = \{1, 2, 3, 4\}$ for $n = 0, 1, 2, 3$. Find $y(n) = x(-n - 2)$.

Step 1 — rewrite the argument : $x(-n - 2) = x(-(n + 2))$

Step 2 — fold : $v(n) = x(-n) = \{4, 3, 2, 1\}$ at $n = -3 \dots 0$

Step 3 — ADVANCE by 2 : $y(n) = v(n + 2) = \{4, 3, 2, 1\}$ at $n = -5 \dots -2$

Check straight from the definition: $y(n)$ is non-zero when $0 \leq -n-2 \leq 3$, that is $-5 \leq n \leq -2$. And $y(-2) = x(0) = 1$, $y(-5) = x(3) = 4$. Agrees.

the common wrong answer – folding then

DELAYING by 2:

$$x(-n + 2) = \{4, 3, 2, 1\} \text{ at } n = -1 \dots 2$$

shifted 4 places wrong

Same four numbers, wrong place on the axis, no marks.

3.4 Even and odd parts

Any signal splits into one mirror-image part and one anti-mirror part, and the split is unique.

$$\begin{aligned} x_e(n) &= \frac{x(n) + x(-n)}{2} & x_e(-n) &= x_e(n) \\ x_o(n) &= \frac{x(n) - x(-n)}{2} & x_o(-n) &= -x_o(n) \\ x(n) &= x_e(n) + x_o(n) & x_o(0) &= 0 \text{ always} \end{aligned}$$

EXAM TIP

The free check is $x_o(0) = 0$. If your odd part has a non-zero value at the origin, you folded wrongly. Takes two seconds.

CHAPTER 03 AT A GLANCE

- Four notations for one signal — functional, tabular, sequence, graphical. In sequence form the **arrow marks $n = 0$** and must be carried everywhere.
- $d(n) = u(n) - u(n - 1)$, and $x(n) = \sum_k x(k)d(n - k)$ — any signal is a row of weighted impulses. That identity is where convolution comes from.
- $x(n - k)$ delays to the right for positive k ; $x(-n)$ mirrors about the origin.
- Fold and shift **do not commute**: rewrite as $x(-(n + k))$ first, then fold, then shift by what is left in the bracket.
- Even part $\frac{x(n) + x(-n)}{2}$, odd part $\frac{x(n) - x(-n)}{2}$, and the odd part is always zero at $n = 0$.

CHAPTER 04

LTI SYSTEMS AND CONVOLUTION

Five yes-or-no questions classify any discrete-time system, and if the answers are linear and time-invariant then one signal — the impulse response $h(n)$ — describes the system completely. This chapter settles the five tests, the two ways of computing a convolution by hand, and the checks that catch a wrong answer before you hand it in.

4.1 The five classification tests

PROPERTY	THE TEST	WHAT FAILS IT
Static (memoryless)	$y(n)$ depends only on $x(n)$ at the same n	Any $x(n-1)$, any sum over past values
Time-invariant	Delay the input by k ; the output must be the old output delayed by k	A coefficient that contains n , or $x(2n)$, or $x(-n)$
Linear	Superposition: $T[ax_1 + bx_2] = aT[x_1] + bT[x_2]$	Squares, products of samples, or an added constant
Causal	$y(n)$ uses no future input	Any $x(n+1)$ term
BIBO stable	Every bounded input gives a bounded output	For an LTI system: $\sum h(n) $ diverging

FOR AN LTI SYSTEM ONLY

Causal $\Leftrightarrow h(n) = 0$ for $n < 0$. **Stable** $\Leftrightarrow \sum |h(n)| < \text{infinity}$. Both tests read straight off $h(n)$ — no trial inputs needed. An FIR system is *always* stable, because a finite sum of finite numbers cannot diverge.

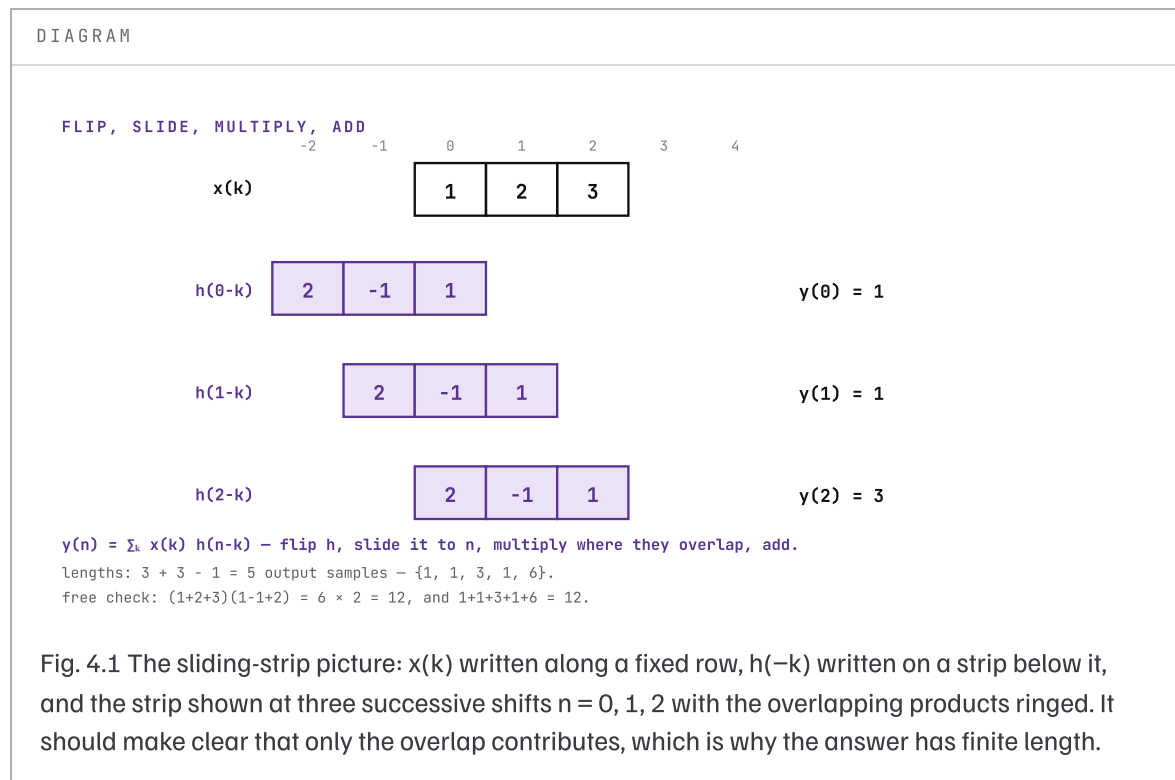
4.2 Where convolution comes from

Break the input into impulses, let the system answer each one, add the answers. Linearity allows the adding, time invariance says every impulse gets the same answer just shifted, and that is the whole derivation.

$$y(n) = \sum_{k=-\infty}^{\infty} x(k)h(n-k) = \sum_k h(k)x(n-k)$$

4.3 Computing one by hand

Two methods. The definition is fold, shift, multiply, add — once per output sample. The faster one for short sequences is the multiplication layout: convolution *is* polynomial multiplication, so line the numbers up and add the columns.



THREE RULES THAT COST NOTHING

Length of the result = $n_1 + n_2 - 1$. **Start index** of the result = start of x + start of h .

Sum check: $\sum y = (\sum x) (\sum h)$.

WORKED EXAMPLE

Two finite sequences. $x(n) = \{1, 2, 3\}$ from $n = 0$ and $h(n) = \{1, -1, 2\}$ from $n = 0$.

length $= 3 + 3 - 1 = 5$, starting at $n = 0 + 0 = 0$

$$y(0) = 1 \cdot 1 = 1$$

$$y(1) = 1(-1) + 2 \cdot 1 = 1$$

$$y(2) = 1 \cdot 2 + 2(-1) + 3 \cdot 1 = 3$$

$$y(3) = 2 \cdot 2 + 3(-1) = 1$$

$$y(4) = 3 \cdot 2 = 6$$

$$y(n) = \{1, 1, 3, 1, 6\}$$

Sum check: sum $y = 1+1+3+1+6 = 12$, and (sum x)(sum h) $= 6 \times 2 = 12$. Agrees.

Now the infinite case, which is the one the paper sets. Take $x(n) = \left(\frac{1}{3}\right)^n u(n)$ and $h(n) = \left(\frac{1}{2}\right)^n u(n)$.

$$\begin{aligned} y(n) &= \sum_{k=0}^n \left(\frac{1}{3}\right)^k \left(\frac{1}{2}\right)^{n-k} \\ &= \left(\frac{1}{2}\right)^n \sum_{k=0}^n \left(\frac{2}{3}\right)^k \\ &= \frac{\left(\frac{1}{2}\right)^n \left[1 - \left(\frac{2}{3}\right)^{n+1}\right]}{1 - \frac{2}{3}} \\ &= 3 \left(\frac{1}{2}\right)^n \left[1 - \left(\frac{2}{3}\right)^{n+1}\right] \\ &= 3 \left(\frac{1}{2}\right)^n - 2 \left(\frac{1}{3}\right)^n \quad \text{for } n \geq 0 \end{aligned}$$

First samples from the formula: 1, 0.8333, 0.5278, 0.3009 — identical to a direct numerical convolution. Note the shape of the answer: **one decaying term per pole**, which is exactly what the z-transform would have given.

4.4 Properties, and reading $h(n)$

- **Commutative** $x * h = h * x$ — you may fold whichever sequence is easier.
- **Associative** — two systems in **cascade** have $h = h_1 * h_2$.
- **Distributive** — two systems in **parallel** have $h = h_1 + h_2$.
- **Step response** = the running sum of $h(n)$. Useful backwards too: $h(n) = s(n) - s(n-1)$.
- Two standard results worth knowing cold: $u * u = (n+1)u(n)$, and $a^n u * b^n u = (a^{n+1} - b^{n+1})/(a - b)$ for $a \neq b$.

EXAM TIP

If a question hands you a step response and asks for the system, difference it:

$h(n) = s(n) - s(n - 1)$. In the continuous version of the same question, differentiate.

That single move is worth seven marks and it turned up in the 2025 paper.

CHAPTER 04 AT A GLANCE

- For an LTI system, $h(n)$ is everything: **causal** $\Leftrightarrow h(n) = 0$ for $n < 0$, **stable** $\Leftrightarrow \sum |h(n)|$ is finite. FIR is always stable.
- $y(n) = \sum_k x(k)h(n - k)$ — break the input into impulses, add the answers.
- Result length = $n_1 + n_2 - 1$, starting at (start of x + start of h).
- Free check every time: $\sum y = (\sum x)(\sum h)$.
- Cascade multiplies the responses ($h_1 * h_2$); parallel adds them ($h_1 + h_2$).
- Step response is the running sum of h ; so $h(n) = s(n) - s(n - 1)$ recovers the system from its step response.

CHAPTER 05

DIFFERENCE EQUATIONS AND THE TOTAL RESPONSE

A difference equation is the system written as arithmetic on past samples. One of these is solved in every single paper. This chapter settles the two decompositions of the answer, how to get the impulse response straight from the coefficients, and how to invert a system.

5.1 The equation and the blocks it stands for

Every discrete-time system is wired from five parts: an adder, a constant multiplier, a signal multiplier, a unit delay z^{-1} , and a unit advance. The wiring diagram and the equation are the same statement.

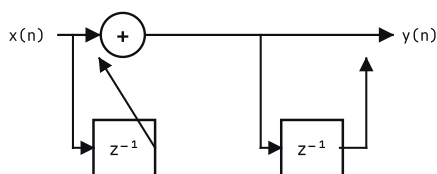
$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$N = 0 \rightarrow$ no feedback $\rightarrow$ FIR, $h(n)$ has finite length

$N > 0 \rightarrow$ feedback $\rightarrow$ IIR, $h(n)$ goes on for ever

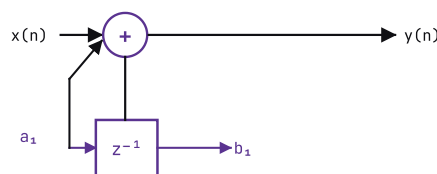
DIAGRAM

DIRECT FORM I - TWO DELAY CHAINS



the feed-forward part and the feedback part each keep their own copy of the past.

DIRECT FORM II - ONE



one delay does both jobs, so half the memory.
that is why form II is the one that gets built.

Fig. 5.1 Direct form I and direct form II for $y(n) - a_1 y(n-1) = b_0 x(n) + b_1 x(n-1)$, drawn side by side with adders, gain triangles and z^{-1} delay boxes labelled. The point of the pair is that form II shares one delay chain and so needs half the memory — that is the mark the realisation question carries.

5.2 Solving by recursion

Set the initial conditions, then turn the handle: $y(0)$, then $y(1)$, then $y(2)$. It always works and it never lies. What it does not give you is a closed formula — so use it to *check* a formula, not to replace one.

5.3 Homogeneous plus particular

- 1 **Homogeneous** — set the right-hand side to zero, try $y(n) = C\lambda^n$, and solve the characteristic equation $\lambda^N + a_1\lambda^{N-1} + \dots + a_N = 0$.
- 2 **Particular** — guess the same shape as the input and fit the constant.
- 3 **Add them**, then fit the unknown constants using $y(0), y(1), \dots$ computed from the difference equation itself.

INPUT $x(n)$	TRY FOR $y_p(n)$
A constant A	K
AM^n	KM^n
An^M	A polynomial $K_0 + K_1n + \dots + KMn^M$
$A \cos(\omega_0 n)$ or $A \sin(\omega_0 n)$	$K_1 \cos(\omega_0 n) + K_2 \sin(\omega_0 n)$
AM^n where M is already a root	KnM^n — multiply by n, as in the repeated-root case

ROOTS TO SOLUTION FORMS

Distinct real roots $\rightarrow \sum C_i \lambda_i^n$. A complex pair $re^{\pm j\theta} \rightarrow r^n [C_1 \cos(\theta n) + C_2 \sin(\theta n)]$. A root repeated m times $\rightarrow r^n [C_0 + C_1 n + \dots + C(m-1)n^{m-1}]$.

5.4 The two decompositions

CUT IT THIS WAY	FIRST PART	SECOND PART
By where the terms come from	Natural — the characteristic roots	Forced — the shape of the input
By what causes it	Zero-input — initial conditions only, input set to zero	Zero-state — input only, system relaxed

THEY ARE NOT THE SAME CUT

Zero-input has the natural form, but the zero-state response *also contains a transient term built from the characteristic roots*. So natural $\neq$ zero-input in general. If the question names one of the four words, give exactly that one.

WORKED EXAMPLE

$y(n) - 0.5y(n-1) = x(n)$, with $x(n) = \left(\frac{1}{4}\right)^n u(n)$ and the system initially relaxed.
Find the complete response.

Homogeneous. $\lambda^n - 0.5\lambda^{n-1} = 0 \rightarrow \lambda = 0.5$
 $y_h(n) = C (0.5)^n$

Particular. input is $(1/4)^n$, so try $y_p(n) = K(1/4)^n$
 $K(1/4)^n - 0.5K(1/4)^{n-1} = (1/4)^n$
 divide by $(1/4)^n$: $K - 0.5K \cdot 4 = 1$
 $K - 2K = 1 \rightarrow K = -1$
 $y_p(n) = -(1/4)^n$

Complete. $y(n) = C(0.5)^n - (0.25)^n$

Initial condition. relaxed means $y(-1) = 0$, so from the equation at $n = 0$:
 $y(0) = 0.5y(-1) + x(0) = 0 + 1 = 1$
 $y(0) = C - 1 = 1 \rightarrow C = 2$

$$y(n) = 2(0.5)^n - (0.25)^n, n \geq 0$$

Check against the raw recursion $y(n) = 0.5y(n-1) + x(n)$:

recursion : 1, 0.75, 0.4375, 0.234375, 0.121094, ...
 formula : 1, 0.75, 0.4375, 0.234375, 0.121094, ... identical

One decaying term from the system pole at 0.5, one from the input pole at 0.25 — that is always the shape.

5.5 Impulse response straight from the equation

THE TWO-LINE RECIPE

Solve the **homogeneous** equation for $h_0(n)$ with the single condition $h_0(0) = 1$ and all earlier values zero. Then combine the numerator coefficients:

$$h(n) = b_0 h_0(n) + b_1 h_0(n-1) + \dots + b_M h_0(n-M).$$

5.6 The inverse system

The inverse undoes the system: cascade the two and you get back exactly what you put in, so $h(n) * h_{\text{inv}}(n) = d(n)$. To build it, swap the roles of x and y in the difference equation — the numerator becomes the denominator.

system : $y(n) - ay(n-1) = x(n)$ $H(z) = \frac{1}{1 - az^{-1}}$
 inverse : $y(n) = x(n) - ax(n-1)$ $H_{\text{inv}}(z) = 1 - az^{-1}$

an IIR system inverts to an FIR one, and the other way round

THE FOLLOW-UP QUESTION

Inverting swaps poles and zeros. So a zero of $H(z)$ *outside* the unit circle becomes a pole of the inverse outside the unit circle, and the inverse is then unstable. The examiner asks “describe its behaviour” precisely to see whether you say that.

CHAPTER 05 AT A GLANCE

- No feedback $\rightarrow$ FIR and always stable; feedback $\rightarrow$ IIR and the roots decide stability.
- Total = **natural + forced = zero-input + zero-state** — two different cuts of the same answer, and zero-state still contains a transient.
- Homogeneous: solve $\lambda^N + a_1\lambda^{N-1} + \dots = 0$; particular: guess the shape of the input; then fit the constants from $y(0)$, $y(1)$ computed by the equation.
- Impulse response: homogeneous solution with $h_0(0) = 1$, then
 $h(n) = b_0h_0(n) + b_1h_0(n-1) + \dots$
- The inverse system swaps x and y — and swaps poles with zeros, so an IIR system inverts to an FIR one.
- Always verify a closed-form answer by running the recursion for three samples. It takes thirty seconds.

CHAPTER 06

THE Z-TRANSFORM AND THE ROC

The z-transform turns convolution into multiplication and difference equations into algebra. But the algebra alone is only half an answer: the same $X(z)$ stands for several different signals, and the region of convergence is what picks between them. This chapter settles the pairs, the properties, and the ROC rules.

6.1 The definition

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

ROC = the set of z for which that sum converges

Each sample becomes one term. The transform is a power series in z^{-1} , and reading the coefficients off a series is itself a way of inverting it.

6.2 The pairs that do almost everything

$x(n)$	$X(z)$	ROC
$\delta(n)$	1	All z
$\delta(n - k)$	z^{-k}	All z except 0 (and ∞ if $k < 0$)
$u(n)$	$\frac{1}{1-z^{-1}}$	$ z > 1$
$a^n u(n)$ — right-sided	$\frac{1}{1-az^{-1}}$	$ z > a $
$-a^n u(-n - 1)$ — left-sided	$\frac{1}{1-az^{-1}}$	$ z < a $
$na^n u(n)$	$a z^{-1} / (1 - a z^{-1})^2$	$ z > a $
$(n + 1)a^n u(n)$	$1 / (1 - a z^{-1})^2$	$ z > a $

ROWS FOUR AND FIVE ARE THE WHOLE POINT

The **same** $X(z)$ with two different ROCs gives two completely different signals — one right-sided, one left-sided, and the left-sided one carries a **minus sign**. An $X(z)$ written without its ROC has not been answered.

6.3 The properties you actually use

delay	$x(n-k)$	$\longleftrightarrow z^{-k} X(z)$
convolution	$x(n) * h(n)$	$\longleftrightarrow X(z) H(z)$
scaling	$a^n x(n)$	$\longleftrightarrow X(z/a)$
ramp weight	$n x(n)$	$\longleftrightarrow -z \frac{dX(z)}{dz}$
fold	$x(-n)$	$\longleftrightarrow X(1/z)$ (ROC inverts too)
initial value	$x(0)$	$= \lim_{z \rightarrow \infty} X(z)$

WHY THE DELAY PROPERTY IS THE WORKHORSE

It turns $y(n) - ay(n-1) = x(n)$ into $Y(z) - az^{-1}Y(z) = X(z)$ in one line, and then $H(z) = \frac{Y(z)}{X(z)}$ falls out by division. Almost every z-transform question in this paper starts with that single move.

6.4 The rules for the ROC

- The ROC is always a **ring** centred on the origin, and it **never contains a pole**.
- **Right-sided** signal $\rightarrow$ the ROC is the *outside* of the outermost pole.
- **Left-sided** signal $\rightarrow$ the ROC is the *inside* of the innermost pole.
- **Two-sided** signal $\rightarrow$ the ROC is an annulus between two poles, or empty.
- **Finite-length** signal $\rightarrow$ the ROC is the whole plane, minus $z = 0$ if it has positive- n terms, minus $z = \infty$ if it has negative- n terms.
- A signal that starts at $n = 0$ has an ROC that includes $z = \infty$. That is exactly the test for causality.

DIAGRAM

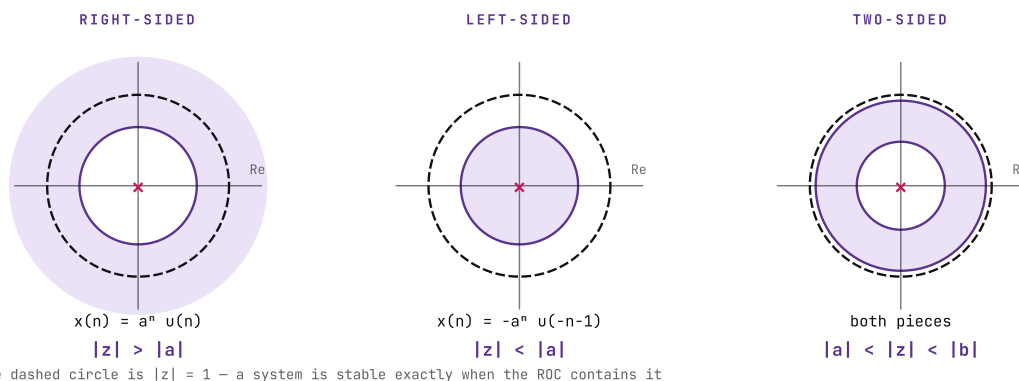


Fig. 6.1 One z-plane with the unit circle drawn and two poles marked at $|z| = 0.5$ and $|z| = 2$, and the three possible rings shaded in turn: the outside ring labelled *causal, not stable*, the middle annulus labelled *two-sided, stable* — it contains $|z| = 1$, and the inner disc labelled *anti-causal, not stable*. One drawing that answers the whole standard question.

KIND OF SIGNAL	ROC	CONTAINS $z = \infty$?	SIGN IN FRONT OF THE TERM
Causal / right-sided $a^n u(n)$	Outside the outermost pole, $ z > a $	Yes	Plus
Anti-causal / left-sided $-a^n u(-n-1)$	Inside the innermost pole, $ z < a $	No	Minus
Two-sided	An annulus between two poles	No	Plus on the inner poles, minus on the outer ones

WORKED EXAMPLE

Find $X(z)$ and its ROC for $x(n) = (0.5)^n u(n) - (2)^n u(-n-1)$.

$$\text{right-sided piece : } (0.5)^n u(n) \leftrightarrow \frac{1}{1 - 0.5z^{-1}}, |z| > 0.5$$

$$\text{left-sided piece : } -(2)^n u(-n-1) \leftrightarrow \frac{1}{1 - 2z^{-1}}, |z| < 2$$

$$X(z) = \frac{1}{1 - 0.5z^{-1}} + \frac{1}{1 - 2z^{-1}}$$

$$\text{ROC} = \text{intersection} = 0.5 < |z| < 2$$

The intersection is not empty, so the signal exists — and the ring contains $|z| = 1$, so the signal is **absolutely summable** and the system is stable. It is not causal: it has values at negative n .

Numerical check at $z = 1$, which lies in the ring: the direct sum gives $2 - 1 = 1$, and the closed form gives $\frac{1}{0.5} + \frac{1}{-1} = 2 - 1 = 1$. They also agree at $z = 1.3 + 0.4j$.

if the second term had been $-(0.3)^n u(-n-1)$:

ROC would need $|z| > 0.5$ AND $|z| < 0.3 \rightarrow \text{EMPTY}$

the z -transform of that signal does not exist

EXAM TIP

The phrase “mention the ROC in all cases” appears in this paper every single year. It is asking for several answers from one algebraic $X(z)$, and most of the marks are in the ROC discussion, not in the partial fractions.

CHAPTER 06 AT A GLANCE

- $X(z) = \sum x(n)z^{-n}$, and the answer is incomplete without its **ROC**.
- $a^n u(n)$ and $-a^n u(-n-1)$ have the **identical** $X(z)$ and differ only in the ROC — note the minus sign on the left-sided one.
- ROC is a ring centred on the origin and never contains a pole: right-sided $\rightarrow$ outside the outermost pole, left-sided $\rightarrow$ inside the innermost.
- Two-sided signals give an annulus, which may be empty — and then no z -transform exists.
- The workhorse property is the delay: $x(n-k) \leftrightarrow z^{-k} X(z)$, which turns a difference equation into algebra in one line.
- Convolution becomes multiplication: $x * h \leftrightarrow X(z)H(z)$.

CHAPTER 07

THE INVERSE Z-TRANSFORM

Going back from $X(z)$ to $x(n)$. Three methods exist, but before any of them there is one question: where is the ROC? Get that wrong and every sign in the answer is wrong. This chapter settles partial fractions done safely, and the three-cases question that is worth five marks a year.

7.1 The three methods

METHOD	USE IT WHEN	WHAT IT GIVES
Partial fractions	$X(z)$ is a ratio of polynomials — almost always	A closed formula for $x(n)$
Power series / long division	You only need the first few samples, or $X(z)$ is not rational	The samples one at a time, no formula
Contour integral / residues	Asked for by name, or the poles are awkward	$x(n) = \text{sum of residues of } X(z)z^{n-1} \text{ inside the contour}$

7.2 Partial fractions, done the safe way

EXPAND $X(z)/z$, NOT $X(z)$

Divide by z first, split into partial fractions, then multiply every term back by z . That leaves each term in the form $\frac{Az}{z-p} = \frac{A}{1-pz^{-1}}$, which is exactly the shape in the table.

Expanding $X(z)$ directly tends to leave a stray constant that is easy to lose.

The alternative, equally safe, is to treat z^{-1} as the variable throughout and match coefficients of 1 , z^{-1} , z^{-2} in turn. Use whichever you have practised — but use the same one every time.

7.3 The ROC picks the sequence

TERM BY TERM

For each pole p , look at where the ROC sits **relative to that pole**. If the ROC is *outside* $|z| = |p|$, that term is causal: $+Ap^n u(n)$. If the ROC is *inside*, that term is anti-causal: $-Ap^n u(-n-1)$. Different poles in the same $X(z)$ can go different ways — that is what makes a two-sided answer.

WORKED EXAMPLE

$H(z) = \frac{3-4z^{-1}}{1-3.5z^{-1}+1.5z^{-2}}$. Give every possible ROC and $h(n)$ in each case, and say which is causal and which is stable.

Step 1

factor the denominator.

$$1 - 3.5z^{-1} + 1.5z^{-2} = (1 - 3z^{-1})(1 - 0.5z^{-1})$$

poles at $z = 3$ and $z = 0.5$

Step 2

partial fractions.

$$H(z) = \frac{A}{1 - 3z^{-1}} + \frac{B}{1 - 0.5z^{-1}}$$

$$3 - 4z^{-1} = A(1 - 0.5z^{-1}) + B(1 - 3z^{-1})$$

constant : $A + B = 3$

$$z^{-1}: -0.5A - 3B = -4$$

solving : $A = 2, B = 1$

$$H(z) = \frac{2}{1 - 3z^{-1}} + \frac{1}{1 - 0.5z^{-1}}$$

Step 3 — two poles cut the plane into three rings, and each ring is a different $h(n)$.

$$|z| > 3$$

both terms causal

$$h(n) = 2(3)^n u(n) + (0.5)^n u(n)$$

causal YES, stable NO (ring excludes $|z| = 1$)

$$0.5 < |z| < 3$$

inside the pole at 3, outside the pole at 0.5

$$h(n) = -2(3)^n u(-n - 1) + (0.5)^n u(n)$$

causal NO, stable YES (ring contains $|z| = 1$)

$$|z| < 0.5$$

both terms anti-causal

$$h(n) = -2(3)^n u(-n - 1) - (0.5)^n u(-n - 1)$$

causal NO, stable NO

The sentence that earns the last mark: **this system cannot be both causal and stable**. Causal forces $|z| > 3$, which excludes the unit circle. Stable forces the annulus, which is two-sided. A causal system is stable only when every pole lies inside $|z| = 1$, and the pole at $z = 3$ does not.

THE MINUS SIGN

Every anti-causal term carries a minus: $-Ap^n u(-n-1)$, never $+Ap^n u(-n-1)$.

Dropping it is the single most common error in this question, and it is worth a mark on its own.

7.4 Long division, and complex poles

For a causal signal, divide in **descending** powers of z so the quotient comes out as $x(0) + x(1)z^{-1} + x(2)z^{-2} + \dots$. For an anti-causal one, divide in **ascending** powers. The coefficients are the samples.

a complex conjugate pole pair at $re^{\pm j\omega_0}$

gives, for a causal signal:

$$x(n) = Ar^n \cos(\omega_0 n + \phi) u(n)$$

the radius r sets how fast it decays,

the angle ω_0 sets how fast it oscillates

EXAM TIP

With a complex pair you are not expected to produce A and ϕ from nowhere. Get r and ω_0 from the poles, write the answer in the $r^n(C_1 \cos + C_2 \sin)$ form, and fit C_1 and C_2 from $x(0)$ and $x(1)$ obtained by long division. Two samples, two unknowns.

CHAPTER 07 AT A GLANCE

- Before inverting anything, find the ROC. It decides the sign and the side of every term.
- Partial fractions: expand $X(z)/z$, split, then multiply back by z — that leaves terms in the $\frac{A}{1-pz^{-1}}$ shape.
- Term by term: ROC outside the pole $\rightarrow +Ap^n u(n)$; ROC inside the pole $\rightarrow -Ap^n u(-n-1)$, **with the minus sign**.
- M poles at distinct radii give **$M+1$** possible ROCs and **$M+1$** different signals from one $X(z)$.
- Long division in descending powers of z gives a causal sequence; ascending powers gives an anti-causal one.
- A complex pole pair $re^{\pm j\omega_0}$ gives $r^n \cos(\omega_0 n + \phi)$ — radius sets the decay, angle sets the oscillation.

CHAPTER 08

SYSTEM FUNCTION, POLES, ZEROS AND STABILITY

$H(z)$ is the difference equation written as one fraction, and everything about the system can be read off where its poles and zeros sit. This chapter settles the two separate tests — causal and stable — and the exact condition under which they can hold at once.

8.1 From difference equation to $H(z)$

$$y(n) + \sum a_k y(n-k) = \sum b_k x(n-k)$$

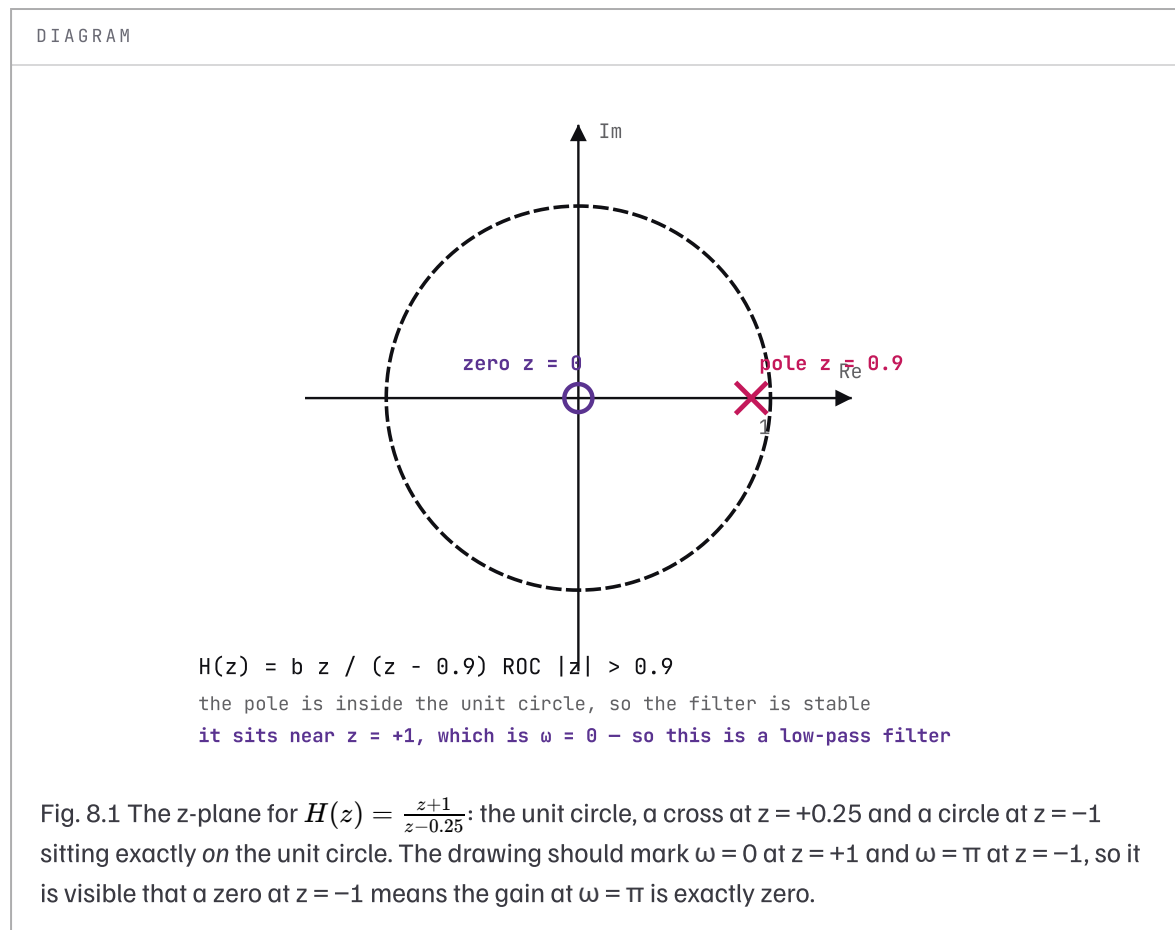
$$H(z) = \frac{Y(z)}{X(z)} = \frac{b_0 + b_1 z^{-1} + \dots + b_M z^{-M}}{1 + a_1 z^{-1} + \dots + a_N z^{-N}}$$

multiply top and bottom by z^N to read poles and zeros directly

POLES AND ZEROS

Zeros are the roots of the numerator — frequencies the system kills. **Poles** are the roots of the denominator — the system's own natural modes. Only the **poles** decide stability. Zeros never do.

8.2 Reading the pole-zero plot



- Pole **inside** the unit circle → a mode that decays. The nearer the origin, the faster.
- Pole **on** the unit circle → a mode that neither grows nor decays. Not BIBO stable.
- Pole **outside** → a mode that grows without limit.
- The **angle** of a pole is the frequency it resonates at; its **radius** is how sharp the resonance is.
- A zero sitting **on** the unit circle at angle ω_0 puts an exact null in the response at ω_0 .

8.3 Causal, stable, or both

PROPERTY	TEST ON THE ROC	TEST ON THE POLES
Causal	ROC is the outside of the outermost pole, and includes $z = \infty$	—
Stable	ROC contains the unit circle $ z = 1$	—
Causal and stable together	—	Every pole strictly inside the unit circle

THESE ARE TWO SEPARATE QUESTIONS

A system can be stable and not causal. A system can be causal and not stable. They coincide only when every pole is inside $|z| = 1$ — then the causal ROC already contains the unit circle, so the causal answer is the stable answer.

WORKED EXAMPLE

$H(z)$ has three poles, at $z = 0.5$, $z = 2$ and $z = 5$. Comment on causality and stability.

$$H(z) = \frac{1}{[(1 - 0.5z^{-1})(1 - 2z^{-1})(1 - 5z^{-1})]}$$

three distinct pole radii $\rightarrow 3 + 1 = \text{FOUR}$ possible ROCs

$$|z| > 5$$

causal NOT stable (unit circle is outside the ring)

$$2 < |z| < 5$$

two-sided NOT stable

$$0.5 < |z| < 2 \text{ two-sided STABLE (contains } |z| = 1)$$

$$|z| < 0.5$$

anti-causal NOT stable

Conclusion to write out: with poles outside the unit circle the system **cannot be both causal and stable**. Exactly one ROC is stable and it is a two-sided one. State which ROC you are assuming before you answer anything else — without it the question has no single answer.

Contrast with $H(z) = \frac{3-4z^{-1}}{1-z^{-1}+0.5z^{-2}}$. The denominator gives $z^2 - z + 0.5 = 0$, so $z = 0.5 \pm j0.5$, a conjugate pair with

$$|z| = \sqrt{0.25 + 0.25} = 0.7071 \quad \text{angle} = \frac{\pi}{4}$$

Both poles are inside the unit circle at the same radius, so there are only **two** ROCs, and the causal one ($|z| > 0.707$) is also the stable one. Here causal and stable give the identical $h(n)$ — a decaying oscillation $(0.707)^n$ at $\omega_0 = \frac{\pi}{4}$.

EXAM TIP

Count the **distinct pole radii** first. M distinct radii means $M + 1$ possible ROCs, and the question “specify all possible regions of convergence” is asking for exactly that many answers. Two poles at the *same* radius count as one radius — that is why a complex pair gives two ROCs, not three.

CHAPTER 08 AT A GLANCE

- $H(z)$ = numerator over denominator; **zeros** from the top, **poles** from the bottom, and only poles decide stability.
- **Causal** $\Leftrightarrow$ ROC is outside the outermost pole. **Stable** $\Leftrightarrow$ ROC contains $|z| = 1$. Two separate tests.
- **Causal and stable together** $\Leftrightarrow$ every pole strictly inside the unit circle. Nothing else.
- M distinct pole radii give $M + 1$ possible ROCs — a complex conjugate pair is one radius, not two.
- A pole's radius sets how fast its mode decays; its angle sets the frequency it resonates at.
- A zero on the unit circle at angle ω_0 puts an exact null in the frequency response at ω_0 .

CHAPTER 09

FREQUENCY RESPONSE

Evaluate $H(z)$ on the unit circle and you get the gain and phase shift the system applies at every frequency. A magnitude-and-phase question appears in every paper. This chapter settles the substitution, the algebraic identity that saves five minutes, and the two-substitution trick for naming the filter type.

9.1 The four Fourier transforms

There are four, and they differ in only two ways: is time continuous or sampled, and does the signal repeat?

	APERIODIC SIGNAL	PERIODIC SIGNAL
Continuous time	CTFT — spectrum continuous and aperiodic	CTFS — spectrum discrete lines, aperiodic
Discrete time	DTFT — spectrum continuous in ω and periodic with period 2π	DFS / DFT — spectrum discrete and periodic : N numbers, then they repeat

THE RULE BEHIND THE GRID

Discrete in one domain $\Leftrightarrow$ periodic in the other. Sampled time $\rightarrow$ the spectrum repeats every 2π . Sampled frequency $\rightarrow$ the time signal repeats every N . Only the DFT is discrete **and** finite in both domains, so only the DFT can be computed.

9.2 The DTFT and the unit circle

$$X(\omega) = \sum_n x(n)e^{-j\omega n} = X(z) \text{ evaluated at } z = e^{j\omega}$$

$$x(n) = \left(\frac{1}{2\pi}\right) \int_{-\pi}^{\pi} X(\omega)e^{j\omega n} d\omega$$

$X(\omega)$ is periodic with period 2π

only $-\pi..pi$ is distinct

WHEN THE SUBSTITUTION IS ILLEGAL

You may replace z by $e^{j\omega}$ only if the ROC **contains** the unit circle. If a pole sits on or outside $|z| = 1$ the DTFT does not exist in the ordinary sense, and the honest answer says so.

9.3 Magnitude and phase

THE IDENTITY THAT SAVES FIVE MINUTES

$|1 - ae^{-j\omega}|^2 = 1 - 2a \cos(\omega) + a^2$. With $a = 0.5$ that is $1.25 - \cos(\omega)$ straight away, with no expanding of $\sin^2$ and $\cos^2$. It appears in every one of these questions.

For a ratio, take magnitudes of top and bottom separately and divide; take angles of top and bottom separately and subtract. Never try to rationalise the whole fraction first.

WORKED EXAMPLE

$y(n) - 0.5y(n-1) = x(n)$. Find the magnitude and phase of the frequency response and say what kind of filter it is.

$$H(z) = \frac{1}{1 - 0.5z^{-1}} \text{ pole at } z = 0.5, \text{ zero at } z = 0$$

$$H(e^{j\omega}) = \frac{1}{1 - 0.5e^{-j\omega}}$$

$$|1 - 0.5e^{-j\omega}|^2 = 1 - 2(0.5)\cos\omega + 0.25 = 1.25 - \cos\omega$$

$$|H(e^{j\omega})| = \frac{1}{\sqrt{1.25 - \cos\omega}}$$

$$\angle H(e^{j\omega}) = -\arctan\left[\frac{0.5\sin\omega}{1 - 0.5\cos\omega}\right]$$

Evaluate at three points:

$\omega = 0$	: $ H = 1/\sqrt{0.25} = 2.0000$	phase = 0
$\omega = \pi/2$	: $ H = 1/\sqrt{1.25} = 0.8944$	phase = -0.4636 rad
$\omega = \pi$	: $ H = 1/\sqrt{2.25} = 0.6667$	phase = 0

Gain falls from 2 at DC to 0.667 at the highest frequency, so this is a **low-pass** system with a DC gain of 2. All three values were confirmed against a direct complex evaluation of $\frac{1}{1-0.5e^{-j\omega}}$.

9.4 Low-pass or high-pass in two substitutions

TWO NUMBERS NAME THE FILTER

The gain at DC is $H(z)$ at $z = +1$, which is just the sum of all the coefficients. The gain at the highest frequency is $H(z)$ at $z = -1$. Bigger at $+1 \rightarrow$ low-pass. Bigger at $-1 \rightarrow$ high-pass. No plotting needed.

example : $H(z) = \frac{1 + z^{-1}}{1 - 0.25z^{-1}}$

$$H(+1) = \frac{1 + 1}{1 - 0.25} = \frac{2}{0.75} = 2.667 \leftarrow \text{DC gain}$$

$$H(-1) = \frac{1 - 1}{1 + 0.25} = \frac{0}{1.25} = 0 \leftarrow \text{exact null}$$

low-pass, and the null is the zero sitting at $z = -1$

INFINITE DC GAIN

If the denominator vanishes at $z = 1$ the DC gain is infinite. Say so plainly: there is a pole on the unit circle at DC, the system is not BIBO stable, and it behaves like an accumulator at low frequencies.

9.5 Steady-state response to a sinusoid

Put a sinusoid into a stable LTI system and, once the transient has died, the *same* sinusoid comes out — only scaled and shifted. This is the single most useful consequence of the frequency response.

in : $x(n) = A \cos(\omega_0 n + \theta)$

out : $y(n) = A|H(\omega_0)| \cos(\omega_0 n + \theta + \text{angle } H(\omega_0))$

the frequency NEVER changes – only amplitude
and phase

EXAM TIP

Worked on the system above at $\omega_0 = \pi/3$: $|H| = 2/\sqrt{3} = 1.1547$ and the phase is $-\pi/6 = -0.5236$ rad. Running the recursion for 400 samples gives exactly that amplitude and lag. If a question gives you a sinusoidal input, you do not need to convolve anything — evaluate H at that one frequency.

CHAPTER 09 AT A GLANCE

- Frequency response = $H(z)$ evaluated at $z = e^{j\omega}$, legal only if the ROC contains the unit circle.
- The DTFT is **continuous in ω and periodic with period 2π** — only $-\pi$ to π is distinct.
- Use $|1 - ae^{-j\omega}|^2 = 1 - 2a \cos \omega + a^2$; take magnitudes and angles of numerator and denominator separately.
- Filter type in two substitutions: $H(+1)$ is the DC gain, $H(-1)$ is the gain at $\omega = \pi$. Bigger at +1 means low-pass.
- A zero on the unit circle gives an exact null; a pole on it gives infinite gain and no BIBO stability.
- Steady state: a sinusoid in gives the **same frequency** out, scaled by $|H|$ and shifted by $\angle H$.

CHAPTER 10

THE DFT AND ITS PROPERTIES

The DFT is the only Fourier transform a computer can actually evaluate: N numbers in, N numbers out. A DFT question appears in every paper, usually a small hand computation with a trap attached. This chapter settles the definition, the free checks, the periodicity trap and circular convolution.

10.1 Why the DFT exists

The DTFT $X(\omega)$ is a continuous function of ω over an infinite set of frequencies. A machine cannot store that. Sample it at N equally spaced points round the unit circle and you get N numbers — the DFT. For a signal of length N or less, nothing at all is lost.

	DTFT $X(\omega)$	DFT $X(k)$
Frequency variable	ω , continuous	$k = 0, 1, \dots, N-1$, discrete
How many values	Infinitely many	Exactly N
Periodic in the frequency variable?	Yes, period 2π	Yes, period N
Signal it implies	The signal as given, running to infinity	The signal repeated every N samples
Computable?	No — it is a formula, not a list	Yes — that is the whole point
Relation	—	$X(k) = X(\omega)$ at $\omega = \frac{2\pi k}{N}$, exactly

10.2 The definition and the twiddle factor

$$W_N = e^{-\frac{j2\pi}{N}}$$

the twiddle factor

$$X(k) = \sum_{n=0}^{N-1} x(n) W_N^{nk} \quad k = 0 \dots N-1$$

$$x(n) = \left(\frac{1}{N}\right) \sum_{k=0}^{N-1} X(k) W_N^{-nk}$$

useful symmetries:

$$W_N^{k+N} = W_N^k \text{ (periodic)}$$

$$W_N^{k+\frac{N}{2}} = -W_N^k$$

half-turn flips the sign

$$W_N^2 = W_{\frac{N}{2}}$$

$$W_4^k = 1, -j, -1, +j$$

FOR $N = 4$, THROW THE TRIGONOMETRY AWAY

$W_4^{nk} = (-j)^{nk}$. Every term is 1, -j, -1 or +j, so a four-point DFT is pure bookkeeping with powers of -j. The paper sets $N = 4$ almost every year for exactly this reason.

10.3 The properties, and the one rule behind them

A DFT of length N treats its N samples as one period of a repeating sequence, so every index is counted **round a ring**: $x((n))_N = x(n \bmod N)$. Shift, and what falls off one end comes back at the other. Fold, and $x((-n))_N = x(N - n)$. Every property is the DTFT property with this wrap-around built in.

COMPUTED · TOOLS/MAKE-FIGURES.PY

The index lives on a ring of $N = 4$ positions. A shift is a rotation; a fold is a reflection through $n = 0$. Nothing falls off the end.

$x(n)$

$x((n-2))_4$ — shifted two places

$x((-n))_4$ — folded

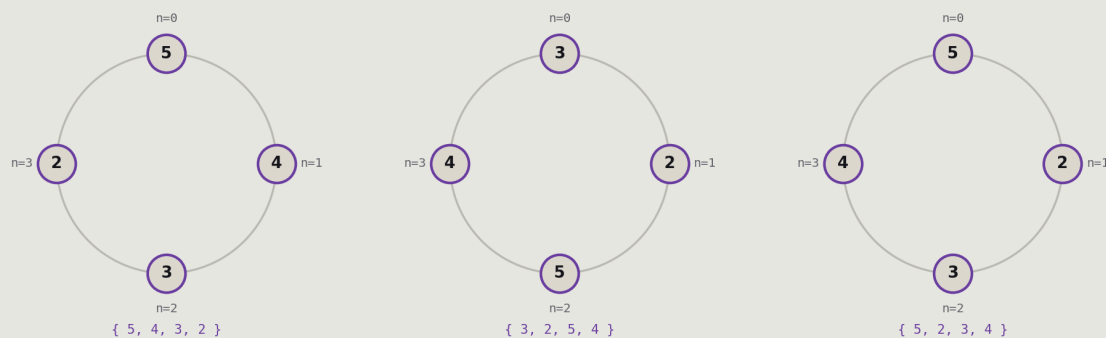


Fig. 10.2 $x(n) = \{5, 4, 3, 2\}$ on a ring; shifted by two it is $\{3, 2, 5, 4\}$; folded it is $\{5, 2, 3, 4\}$ — $x(0)$ stays, the rest reverse.

PROPERTY	SEQUENCE	DFT
Periodicity	$x(n + N) = x(n)$	$X(k + N) = X(k)$
Circular time shift	$x((n - m))_N$	$W_N^{km} X(k)$
Circular frequency shift	$x(n)e^{\frac{j2\pi ln}{N}}$	$X((k - l))_N$
Time reversal (fold)	$x((-n))_N = x(N - n)$	$X(N - k)$
Conjugate	$x(n)$	$X(N - k)$
Real x(n)	—	$X(N - k) = X(k)$: X even, phase odd about N/2
Real and even	$x(n) = x(N - n)$	real: $\sum x(n) \cos\left(\frac{2\pi kn}{N}\right)$
Real and odd	$x(n) = -x(N - n)$	imaginary: $-j \sum x(n) \sin\left(\frac{2\pi kn}{N}\right)$
Circular convolution	$x_1 (*) x_2$	$X_1(k)X_2(k)$
Multiplication	$x_1(n)x_2(n)$	$(1/N) X_1(k) (*) X_2(k)$
Circular correlation	$\sum x(n)y((n - l))_N$	$X(k)Y(k)$
Parseval	$\sum x(n) ^2$	$(\frac{1}{N}) \sum X(k) ^2$

THE PROOF THAT GETS ASKED

Put $N - k$ into the definition: $X(N - k) = \sum x(n)e^{-\frac{j2\pi(N-k)n}{N}} = \sum x(n)e^{+\frac{j2\pi kn}{N}}$
because $e^{-j2\pi n} = 1$, and for real $x(n)$ that is the conjugate of the original sum. So
 $X(N - k) = X(k)$, $X(0)$ is real, and for even N so is $X(\frac{N}{2})$.

WORKED EXAMPLE

The first five points of an 8-point DFT of a real sequence are $\{0.25, 0.125 - j0.3018, 0, 0.125 - j0.0518, 0\}$. Find the other three.

real $x(n) \rightarrow X(k) = X^*(8 - k)$ $X(5) = X^*(3) = 0.125 + j0.0518$ $X(6) = X^*(2) = 0$ $X(7) = X^*(1) = 0.125 + j0.3018$ check : $X(0)$ and $X(4)$ are real, as they must be

Transforming the completed eight values back gives a real sequence — that is the check that the three points were filled in correctly (confirmed with numpy).

COMPUTED · TOOLS/MAKE-FIGURES.PY

Given $X(0) \dots X(4)$ of a real 8-point sequence, $X(N-k) = X^*(k)$ supplies the rest — and transforming back really does give a real $x(n)$.

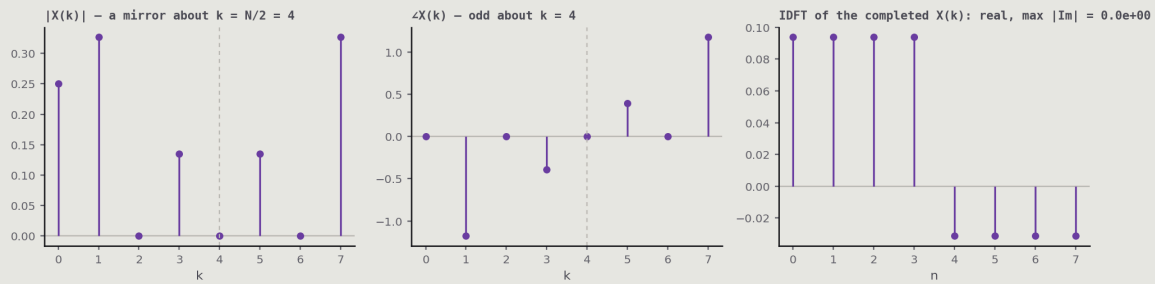


Fig. 10.3 $|X(k)|$ mirrors about $k = 4$ and the phase flips sign; the IDFT of the completed values is real to machine precision.

CIRCULAR, EVERY TIME

Asked for the DFT of $x(n-1)$ “by the shift property” with $x = \{1, 2, 3, 4\}$, the sequence that property applies to is $\{4, 1, 2, 3\}$, not $\{0, 1, 2, 3\}$. Asked for the sequence whose DFT is $X(k)$, the answer is the fold $x((-n))_N = \{1, 4, 3, 2\}$ — $x(0)$ stays put.

10.4 The periodicity trap

WORKED EXAMPLE

$x(n) = \{1, 2, 3, 4\}$ for $n = 0..3$. Compute the DFT, then state $X(4)$ through $X(7)$.

$$N = 4, \quad \text{so} \quad W_4^{nk} = (-j)^{nk}$$

$$X(0) = 1 + 2 + 3 + 4 = 10$$

$$X(1) = 1 + 2(-j) + 3(-j)^2 + 4(-j)^3 = 1 - 2j - 3 + 4j = -2 + 2j$$

$$X(2) = 1 + 2(-j)^2 + 3(-j)^4 + 4(-j)^6 = 1 - 2 + 3 - 4 = -2$$

$$X(3) = 1 + 2(-j)^3 + 3(-j)^6 + 4(-j)^9 = 1 + 2j - 3 - 4j = -2 - 2j$$

$$X(k) = \{10, -2 + 2j, -2, -2 - 2j\}$$

Two free checks. $X(0) = 1+2+3+4 = 10 \checkmark$. And $x(n)$ is real, so $X(3)$ must be the conjugate of $X(1)$: $-2-2j$ against $-2 + 2j \checkmark$. $X(2)$ sits at $N/2$ and must be purely real — it is $\checkmark$.

$$X(4) = X(0) = 10$$

$$X(5) = X(1) = -2 + 2j$$

$$X(6) = X(2) = -2$$

$$X(7) = X(3) = -2 - 2j$$

Confirmed against numpy: the 4-point DFT is exactly $[10, -2 + 2j, -2, -2 - 2j]$, and the inverse transform returns $1, 2, 3, 4$.

BEING ASKED FOR EIGHT VALUES DOES NOT MAKE IT AN 8-POINT DFT

A 4-point DFT has exactly **four** distinct values and then repeats for ever. An 8-point DFT would need eight samples, or an explicit instruction to zero-pad to 8 — and that would give genuinely different numbers, not repeats. This exact trap has been set twice in three years.

10.5 Circular convolution, and zero padding

Multiplying two DFTs does **not** give the ordinary convolution. It gives the circular one, where the index wraps round instead of running off the end. The tails that fall off the end come back at the beginning and add.

$$\text{circular : } y(n) = \sum_{k=0}^{N-1} x(k)h((n-k) \bmod N)$$

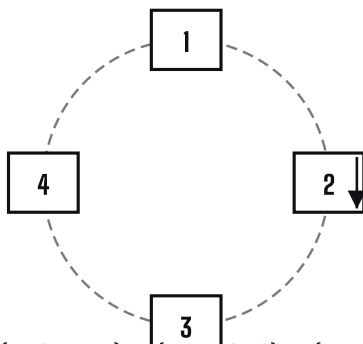
to get the LINEAR answer from a circular convolution,

zero-pad both signals to length $L \geq N + M - 1$

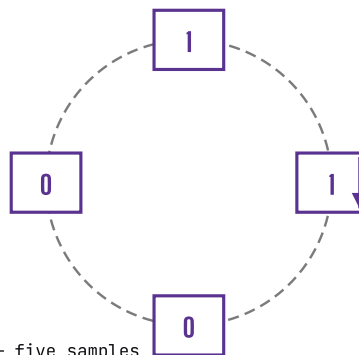
DIAGRAM

CIRCULAR CONVOLUTION - $N = 4$

$x(n)$



$h(-n) \bmod 4$



linear: $\{1, 2, 3, 4\} * \{1, 1, 0, 0\} = \{1, 3, 5, 7, 4\}$ - five samples

circular: the 5th wraps onto the 1st - $4 + 1 = 5$

$\{5, 3, 5, 7\}$

pad both to $3 + 2 - 1 = 5$ points first and the two agree - that is how the FFT does linear convolution.

Fig. 10.2 Two concentric rings of four samples each, the inner one reversed and rotated one step at a time, with the four products for $n = 0$ marked. Alongside it, the 5-point linear result $\{1, 3, 5, 7, 4\}$ with its last sample folded back onto the first to give $\{5, 3, 5, 7\}$ — the wraparound made visible.

WORKED EXAMPLE

$x(n) = \{1, 2, 3, 4\}$ and $h(n) = \{1, 1, 0, 0\}$, both of length $N = 4$. Do the circular convolution and compare it with the linear one.

$$\begin{aligned} y(0) &= x_0h_0 + x_1h_3 + x_2h_2 + x_3h_1 = 1 + 0 + 0 + 4 = 5 \\ y(1) &= x_0h_1 + x_1h_0 + x_2h_3 + x_3h_2 = 1 + 2 + 0 + 0 = 3 \\ y(2) &= x_0h_2 + x_1h_1 + x_2h_0 + x_3h_3 = 0 + 2 + 3 + 0 = 5 \\ y(3) &= x_0h_3 + x_1h_2 + x_2h_1 + x_3h_0 = 0 + 0 + 3 + 4 = 7 \\ \text{circular } y(n) &= \{5, 3, 5, 7\} \end{aligned}$$

Now the linear convolution of $\{1, 2, 3, 4\}$ with $\{1, 1\}$:

$$\begin{aligned} \text{linear} &= \{1, 3, 5, 7, 4\} \text{ length } 4 + 2 - 1 = 5 \\ &\text{the 5th sample has nowhere to go in a 4-point} \\ &\text{circle,} \\ &\text{so it wraps round and adds to the first:} \\ \{1 + 4, 3, 5, 7\} &= \{5, 3, 5, 7\} \qquad \text{matches exactly} \end{aligned}$$

Pad both to length 5 or more and the wraparound disappears: the circular answer then equals the linear one. Both results were confirmed numerically.

WHAT ZERO PADDING DOES AND DOES NOT DO

It **does** put the DFT points closer together, so the plot looks smoother, and it **does** let circular convolution give the linear answer. It **does not** improve frequency resolution. Resolution needs a longer *record*, not more zeros. Bin spacing is $F_k = \frac{kF_s}{N}$.

10.6 Filtering a long signal: overlap-add and overlap-save

A DFT convolves blocks of a fixed size, so a long input is cut into blocks and each block is filtered by an N -point DFT, $N \geq L + M - 1$. Two ways to handle the joins, and the paper asks for both.

	OVERLAP-ADD	OVERLAP-SAVE
Input blocks	L samples, no overlap	N samples, overlapping by $M-1$
Output of each block	$N = L + M - 1$ samples	N samples, first $M-1$ wrong
At the join	add the $M-1$ overlapping outputs	throw away the first $M-1$ outputs, keep the rest
First block	as it is	starts with $M-1$ zeros

WORKED EXAMPLE

Class notebook: $h = \{2, 2, 1\}$, $x = \{3, 0, -2, 0, 2, 1, 0, -2, -1, 0\}$, 8-point DFTs.

direct answer : $x * h = \{6, 6, -1, -4, 2, 6, 4, -3, -6, -4, -1, 0\}$ overlap-add : $L = 8 - 3 + 1 = 6$ block 1 $\{3, 0, -2, 0, 2, 1\} \rightarrow \{6, 6, -1, -4, 2, 6, 4, 1\}$ at $n = 0..7$ block 2 $\{0, -2, -1, 0\} \rightarrow \{0, -4, -6, -4, -1, 0, 0, 0\}$ at $n = 6..13$ $y(6) = 4 + 0 = 4$, $y(7) = 1 - 4 = -3$ the overlap, added overlap-save : blocks of 8, overlapping by 2 block 1 $\{0, 0, 3, 0, -2, 0, 2, 1\} \rightarrow \{4, 1 \mid 6, 6, -1, -4, 2, 6\}$ discard 4, 1 block 2 $\{2, 1, 0, -2, -1, 0, 0, 0\} \rightarrow \{4, 6 \mid 4, -3, -6, -4, -1, 0\}$ discard 4, 6 both give the direct answer exactly

COMPUTED · TOOLS/MAKE-FIGURES.PY

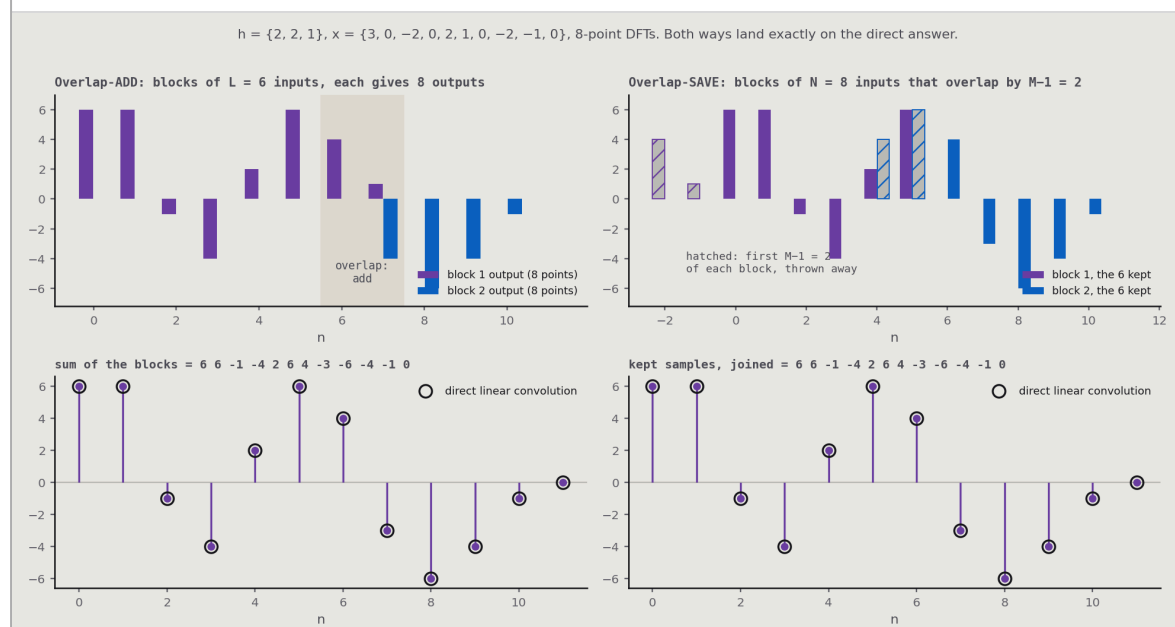


Fig. 10.4 Both methods on the class example: the block outputs, the overlap added or the first two samples thrown away, and the joined result sitting on the direct linear convolution.

CHAPTER 10 AT A GLANCE

- $X(k) = \sum x(n)W_N^{nk}$ with $W_N = e^{-j\frac{2\pi}{N}}$; for $N = 4$ that is just powers of $-j$.
- Two free checks on every DFT: $X(0) = \sum x(n)$, and for real $x(n)$ $X(N - k) = \text{conj}(X(k))$ with $X(\frac{N}{2})$ purely real.
- The DFT is **periodic in k with period N** — asking for $X(4)..X(7)$ of a 4-point DFT is asking for $X(0)..X(3)$ again.
- Multiplying two DFTs gives **circular** convolution; the tail wraps round and adds to the front.
- Zero-pad both signals to $L \geq N + M - 1$ and circular convolution equals the linear answer.
- Zero padding adds dots, not resolution. Resolution needs a longer record; bin spacing is $\frac{F_s}{N}$.

CHAPTER 11

THE FFT

The FFT is not a different transform — it is the same DFT computed by splitting it in half over and over. This is the most reliably examined topic in the paper and the one that moves around most. This chapter settles both decimations, the counting formulas, bit reversal, and what changes for a different radix.

11.1 What direct computation costs

direct DFT:

$$\text{complex multiplications} = N^2$$

$$\text{complex additions} = N(N - 1)$$

radix-2 FFT:

$$\text{complex multiplications} = \left(\frac{N}{2}\right) \log_2 N$$

$$\text{complex additions} = N \log_2 N$$

$$\text{stages} = \log_2 N$$

$$\text{butterflies per stage} = \frac{N}{2}$$

THE TWO SYMMETRIES THAT MAKE IT WORK

$W_N^{k+N} = W_N^k$ (periodicity) and $W_N^{k+\frac{N}{2}} = -W_N^k$ (a half-turn flips the sign). Between them, half of every product has already been computed — that is the entire saving, and it is the sentence to open the answer with.

11.2 Radix-2 decimation in time

Split $x(n)$ into its even-indexed and odd-indexed samples. Each half is an $N/2$ -point DFT, and the two are recombined with one twiddle multiplication per output pair. Repeat the split until only 2-point DFTs are left.

$$X(k) = G(k) + W_N^k H(k)$$

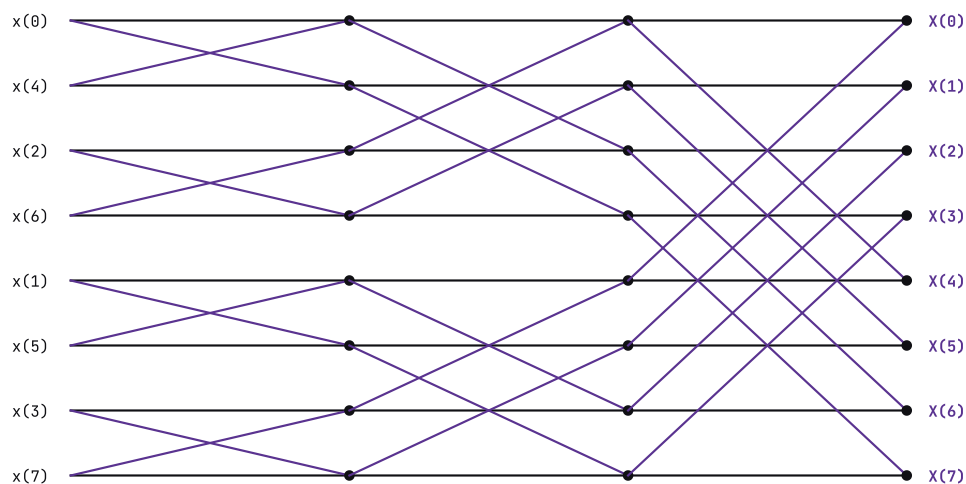
$$X\left(k + \frac{N}{2}\right) = G(k) - W_N^k H(k) \quad k = 0 \dots \frac{N}{2} - 1$$

$$G(k) = \text{DFT of the even-indexed samples}$$

$$H(k) = \text{DFT of the odd-indexed samples}$$

DIAGRAM

DECIMATION IN TIME – input bit-reversed, output in order
stage 1 stage 2 stage 3



$N = 8$, so $\log_2 8 = 3$ stages of $N/2 = 4$ butterflies each
 $(N/2) \log_2 N = 12$ complex multiplies against $N^2 = 64$ for the direct DFT – an 81% saving

Fig. 11.2 The full 8-point DIT flow graph: three stages, four butterflies each, inputs labelled in bit-reversed order $x(0)$, $x(4)$, $x(2)$, $x(6)$, $x(1)$, $x(5)$, $x(3)$, $x(7)$ and outputs $X(0)$.. $X(7)$ in natural order. The twiddles per stage must be shown: W_8^0 in stage 1; W_8^0 and W_8^2 in stage 2; W_8^0 , W_8^1 , W_8^2 and W_8^3 in stage 3.

11.3 Bit reversal and in-place working

Splitting into evens and odds over and over leaves the input in bit-reversed order. Write each index in $\log_2 N$ bits and read the bits backwards.

$N = 8$, three bits:

$0 = 000 \rightarrow 000 = 0$	$4 = 100 \rightarrow 001 = 1$
$1 = 001 \rightarrow 100 = 4$	$5 = 101 \rightarrow 101 = 5$
$2 = 010 \rightarrow 010 = 2$	$6 = 110 \rightarrow 011 = 3$
$3 = 011 \rightarrow 110 = 6$	$7 = 111 \rightarrow 111 = 7$

DIT input order : $x(0)$ $x(4)$ $x(2)$ $x(6)$ $x(1)$ $x(5)$ $x(3)$ $x(7)$

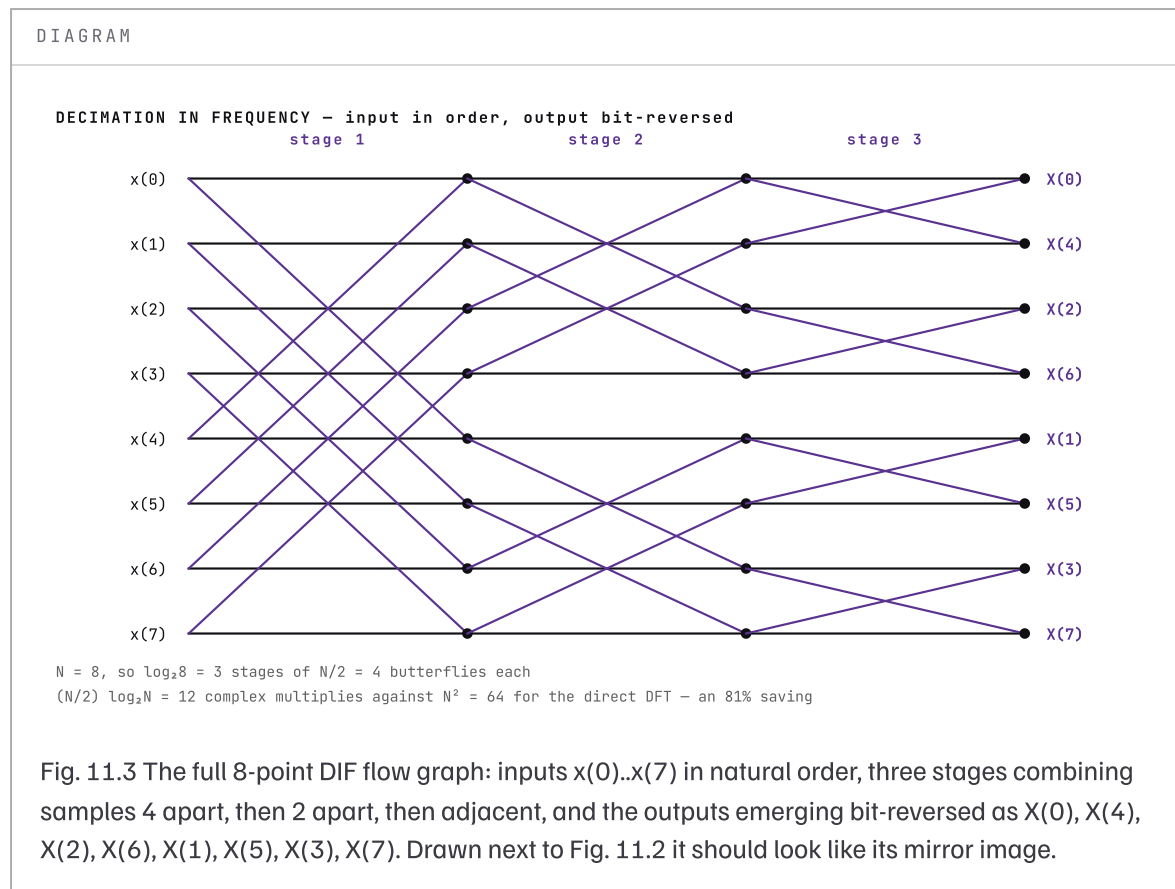
DIT output order: $X(0)$ $X(1)$ $X(2)$ $X(3)$ $X(4)$ $X(5)$ $X(6)$ $X(7)$

IN-PLACE COMPUTATION

Each butterfly reads two values, computes two results, and writes them back over the same two locations. No second array is needed — total storage is about $5N/2$ words ($2N$ for the complex data plus $N/2$ for the twiddles). Bit reversal is the price paid for working in place, and it is worth a mark to say so.

11.4 Radix-2 decimation in frequency

The same saving from the other end: split the **output** into even-numbered and odd-numbered frequencies instead of splitting the input into even and odd samples. Stage 1 combines samples that are $N/2$ apart, stage 2 combines samples $N/4$ apart, and so on down to adjacent pairs.



	DECIMATION IN TIME (DIT)	DECIMATION IN FREQUENCY (DIF)
What is split	The input into even and odd samples	The output into even and odd frequencies
Butterfly	$A = a + W^r b, B = a - W^r b$	$A = a + b, B = (a - b)W^r$
Where the twiddle sits	Before the add and subtract	After the add and subtract
Input order	Bit-reversed	Natural
Output order	Natural	Bit-reversed
Stage 1 pairs samples	Adjacent	$N/2$ apart
Operation count	$(\frac{N}{2}) \log_2 N$ mults, $N \log_2 N$ adds	Exactly the same
In place?	Yes	Yes

BIT-REVERSING THE WRONG END

DIT takes bit-reversed **input**. DIF gives bit-reversed **output**. They are mirror images, and swapping them is the commonest way to lose marks on a flow graph that is otherwise correct.

11.5 Counting the operations

WORKED EXAMPLE

Compare direct computation with the radix-2 FFT, at $N = 8$ and at $N = 1024$.

$N = 8$ $\log_2 N = 3$ stages , $N/2 = 4$ butterflies per stage

direct : mults = N^2 = 64 adds = $N(N-1)$ = 56

FFT : mults = $(8/2)*3$ = 12 adds = $8*3$ = 24

speed-up = $64/12 = 5.33$ times fewer multiplications

saving = $(64 - 12)/64 = 81.25$ %

$N = 1024$ $\log_2 N = 10$ stages

direct : mults = 1,048,576 adds = 1,047,552

FFT : mults = $512*10$ = 5,120 adds = $1024*10$ = 10,240

speed-up = $1,048,576 / 5,120 = 204.8$ times

saving = 99.51 %

Say this about the saving: it **grows with N**. About 5× at $N = 8$, 64× at $N = 256$, 114× at $N = 512$, 205× at $N = 1024$. An N^2 problem has become an $N \cdot \log N$ one, which is the whole reason the algorithm was worth inventing.

$N = 256$, the version the 2024 paper set:

direct : 65,536 multiplications

radix-2 : 1,024 improvement 98.44 % (64x)

split-radix : 683 $\sim (N/3)\log_2 N$ improvement 98.96 % (96x)

11.6 The inverse FFT

DFT : $X(k) = \sum x(n) W_N^{+nk}$

IDFT : $x(n) = (1/N) \sum X(k) W_N^{-nk}$

so the structure is IDENTICAL -- only two things change:

1. conjugate every twiddle ($W^r \rightarrow W^{-r}$)
2. divide the output by N

or reuse the forward FFT untouched:

$$x(n) = (1/N) * \text{conj}(\text{FFT}(\text{conj}(X(k))))$$

EXAM TIP

The operation count for the IFFT is the **same** as for the FFT. If a question asks for the DIT FFT applied to the inverse DFT — as the 2024 paper did — draw the ordinary butterfly, write W^{-r} on it, add the $1/N$ at the output, and quote the same counts.

11.7 Other radices

ALGORITHM	COMPLEX MULTIPLICATIONS	NEEDS N TO BE
Direct DFT	N^2	Anything
Radix-2	$\left(\frac{N}{2}\right) \log_2 N$	A power of 2
Radix-4	$\left(\frac{3N}{8}\right) \log_2 N$	A power of 4
Split-radix	$\approx \left(\frac{N}{3}\right) \log_2 N$	A power of 2
Radix-r, general	$N(r-1)/r \times \log_r N$	A power of r

radix-3 example, $N = 9$:

stages = $\log_3 9 = 2$

butterflies = $N/3 = 3$ per stage, each a 3-point DFT

multiplications = $9 * (2/3) * 2 = 12$ against $N^2 = 81$

speed-up = $81/12 = 6.75$ times

input order: digit reversal in BASE 3

$x(0) \ x(3) \ x(6) \ x(1) \ x(4) \ x(7) \ x(2) \ x(5) \ x(8)$

WHY OTHER RADICES EXIST AT ALL

Radix-2 needs N to be a power of two. If your record is naturally 9 or 27 samples long, padding up to 16 or 32 changes the frequency grid. Radix-3 keeps the natural length. Every radix is the same divide-and-conquer idea with a different number, and every one of them can be computed in place.

GOERTZEL: WHEN YOU WANT ONE BIN, NOT ALL OF THEM

Treat $X(k)$ as the output at $n = N$ of a filter with a pole at the bin frequency:

$$v(n) = x(n) + 2 \cos\left(\frac{2\pi k}{N}\right) v(n-1) - v(n-2), \text{ then } X(k) = v(N) - W_N^k v(N-1).$$

About N real multiplications per bin and no twiddle table, so it beats the FFT when you need fewer than about $\log_2 N$ bins — tone detection, DTMF.

COMPUTED · TOOLS/MAKE-FIGURES.PY

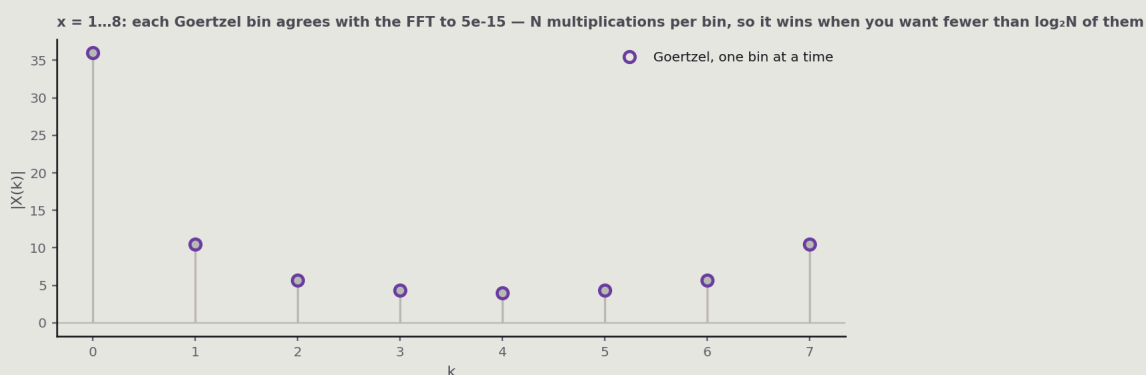


Fig. 11.4 Goertzel on $x = 1 \dots 8$, one bin at a time, landing on the FFT values to machine precision.

CHAPTER 11 AT A GLANCE

- The FFT computes the same DFT; the saving comes from $W_N^{k+\frac{N}{2}} = -W_N^k$ and $W_N^{k+N} = W_N^k$.
- Radix-2 costs $\left(\frac{N}{2}\right) \log_2 N$ multiplications and $N \log_2 N$ additions, in $\log_2 N$ stages of $N/2$ butterflies — against N^2 and $N(N-1)$ direct.
- At $N = 8$ that is 12 multiplications instead of 64 (5.3×); at $N = 1024$ it is 5,120 instead of 1,048,576 (205×).
- **DIT** splits the input, twiddle before the add, **bit-reversed input**. **DIF** splits the output, twiddle after the add, **bit-reversed output**. Identical cost.
- Both work **in place** in about $5N/2$ words — bit reversal is the price paid for that.
- The IFFT is the same flow graph with conjugated twiddles and a final divide by N ; the operation count is unchanged.

CHAPTER 12

FIR FILTER DESIGN

The first design unit. You are told what the filter must do and you find the coefficients. FIR filters are the ones that can have a perfectly straight phase, and the class has reached this unit, so expect a question on it. This chapter settles the four symmetric families, the window method with every ideal response, the window table, how to pick the length, the Kaiser recipe, frequency sampling, and the two special filters.

12.1 Why FIR, and the specification

No feedback, so $h(n)$ has M taps, the only poles are at $z = 0$, and the filter is always stable. The big gain is **linear phase**: every frequency is delayed by the same $\frac{M-1}{2}$ samples, so a waveform keeps its shape. The cost is length: a sharp cutoff needs many taps.

$$\begin{aligned}
 1 - \delta_1 &\leq |H(\omega)| \leq 1 + \delta_1 & 0 \leq \omega \leq \omega_p & \text{(passband)} \\
 |H(\omega)| &\leq \delta_2 & \omega_s \leq \omega \leq \pi & \text{(stopband)} \\
 \Delta\omega &= \omega_s - \omega_p & & \text{(transition band)}
 \end{aligned}$$

COMPUTED · TOOLS/MAKE-FIGURES.PY

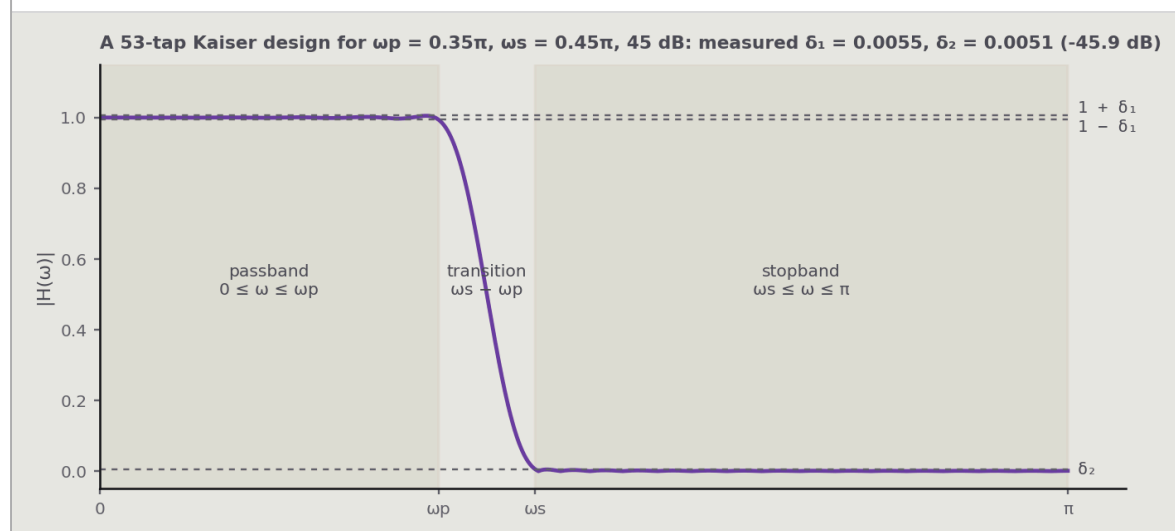


Fig. 12.1 The specification, read off a real Kaiser design: passband ripple, stopband level and transition band, all measured.

WHY A PERFECT FILTER IS IMPOSSIBLE

Paley–Wiener: a causal filter cannot have $|H| = 0$ over a whole band, and cannot jump from 1 to 0. So every ideal brick wall is non-causal, and design means approximation: a transition band and some ripple are the price of a finite, causal filter. Causal means $h(n) = 0$ for $n < 0$, or: finite zeros no more than finite poles.

12.2 Linear phase: the four families

$$h(n) = \pm h(M-1-n), \quad \text{delay } \tau = \frac{M-1}{2}, \quad H(e^{j\omega}) = e^{-j\omega\tau} H_r(\omega) \quad (\text{or } j e^{-j\omega\tau} H_r \text{ if antisymmetric})$$

TYPE	SYMMETRY	M	FORCED ZEROS	USE IT FOR
1	symmetric	odd	none	anything
2	symmetric	even	$H(\pi) = 0$	LP, BP only
3	antisymmetric	odd	$H(0) = H(\pi) = 0$	BP, differentiator, Hilbert
4	antisymmetric	even	$H(0) = 0$	HP, BP, differentiator, Hilbert

COMPUTED · TOOLS/MAKE-FIGURES.PY

Linear phase needs symmetry about the centre tap. Which of the four families you can use depends on the filter you want.

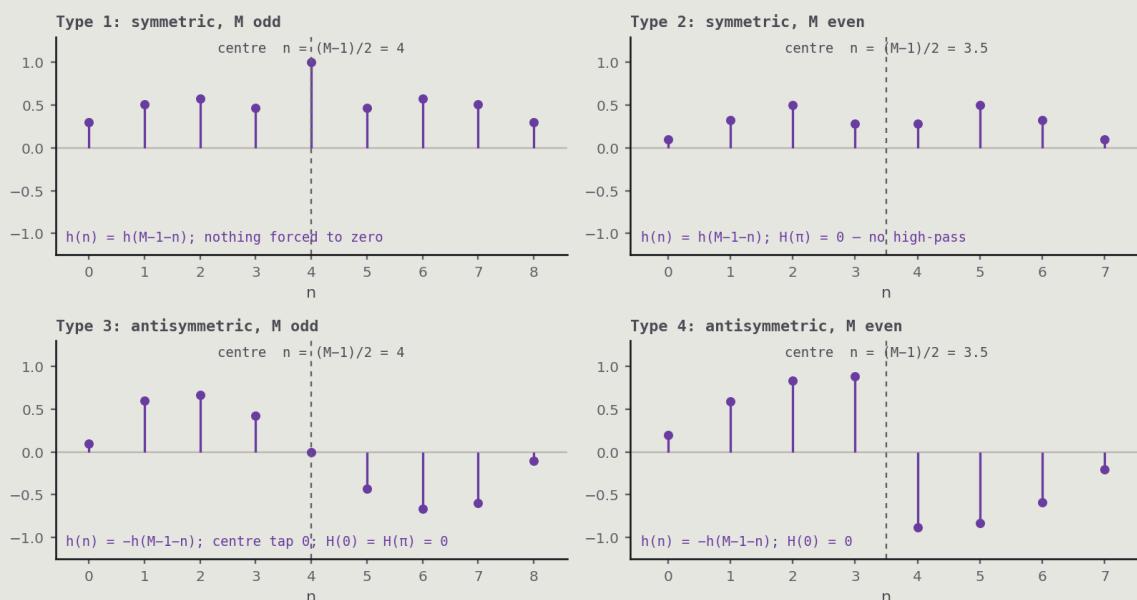


Fig. 12.2 The four families with the centre of symmetry marked. The forced zeros written on each panel follow from the symmetry alone.

$$\text{Type 1: } H_r(\omega) = h(\tau) + 2 \sum_{n=0}^{\tau-1} h(n) \cos \omega(\tau - n), \quad \angle H = -\omega\tau \quad (+\pi \text{ where } H_r < 0)$$

EVEN LENGTH CANNOT BE HIGH-PASS

A symmetric filter of even length has $H(\pi) = 0$ exactly. If you are asked for a linear-phase high-pass with even M, say it needs Type 4 (antisymmetric) or an odd M — that sentence is the mark.

WHERE THE ZEROS SIT

Symmetry gives $H(z) = z^{-M-1} H\left(\frac{1}{z}\right)$, so zeros come in reciprocal pairs; real coefficients add conjugates. Off the unit circle they come in groups of four, z_0 , $1/z_0$, z_0^* , $1/z_0^*$; on it, in conjugate pairs; at $z = \pm 1$, alone. So a linear-phase FIR is never minimum phase unless every zero is on the circle.

COMPUTED · TOOLS/MAKE-FIGURES.PY

Zeros of the 11-tap low-pass, $\omega_c = \pi/2$ (rectangular window). Symmetric $h(n)$ means $H(z) = z^{-(M-1)} H(1/z)$: every zero at z comes with one at $1/z$, and real coefficients add the conjugate of each.

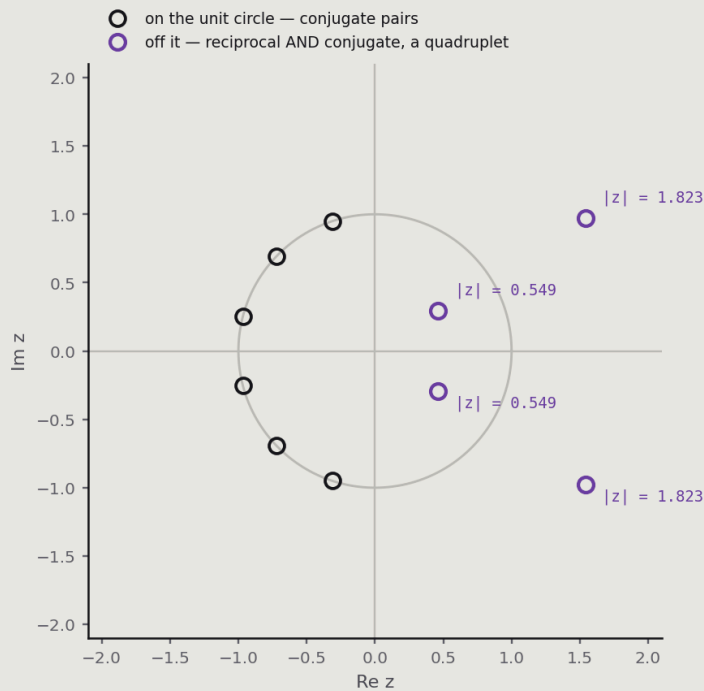


Fig. 12.3 The zeros of the 11-tap low-pass designed in 12.4, from np.roots: six on the circle in conjugate pairs, four in a group at $|z| = 0.549$ and $1/0.549$.

12.3 The window method

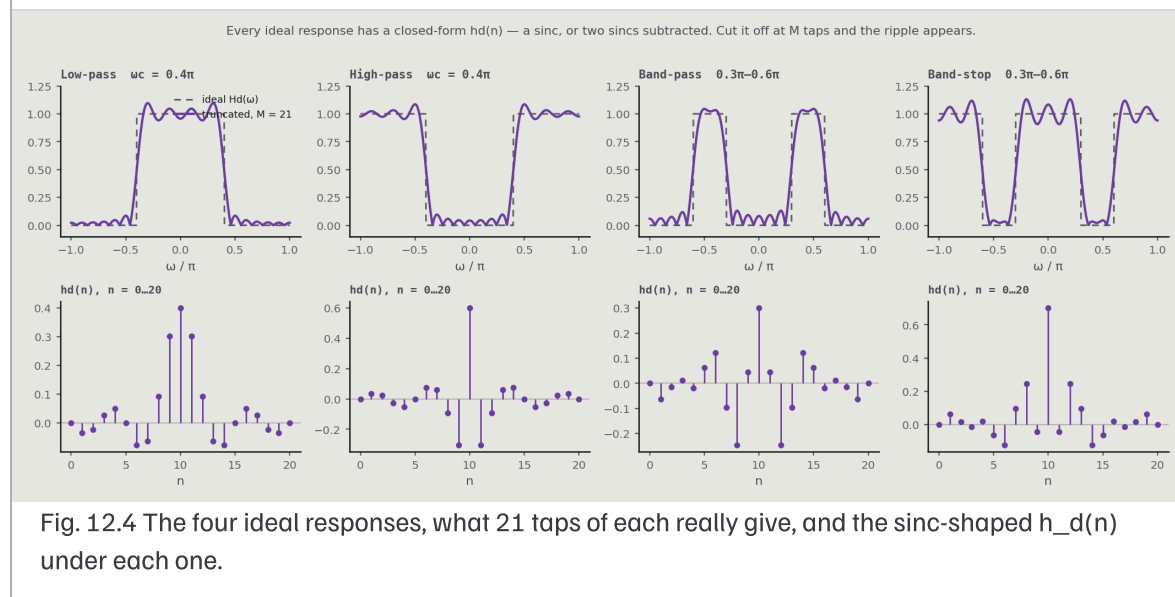
- 1 Write $H_d(e^{j\omega})$ with the delay $e^{-j\omega\tau}$, $\tau = \frac{M-1}{2}$.
- 2 Invert it: $h_d(n) = \left(\frac{1}{2\pi}\right) \int H_d e^{j\omega n} d\omega$ — a sinc.
- 3 Cut it: $h(n) = h_d(n)\omega(n)$ for $0 \leq n \leq M-1$.
- 4 Write $H(z)$, pairing the mirror taps.
- 5 Make a table of $|H(\omega)|$ at $\omega = 0, 0.1\pi, \dots, \pi$ (and in dB).

FILTER	$H_D(N), N \neq T$	$H_D(T)$
Low-pass w_c	$\frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$	$\frac{\omega_c}{\pi}$
High-pass w_c	$\frac{[\sin(\pi(n-\tau)) - \sin(\omega_c(n-\tau))]}{\pi(n-\tau)}$	$1 - \frac{\omega_c}{\pi}$
Band-pass $w_1..w_2$	$\frac{[\sin(w_2(n-\tau)) - \sin(w_1(n-\tau))]}{\pi(n-\tau)}$	$\frac{w_2 - w_1}{\pi}$
Band-stop $w_1..w_2$	$\frac{[\sin(w_1(n-\tau)) - \sin(w_2(n-\tau)) + \sin(\pi(n-\tau))]}{\pi(n-\tau)}$	$1 - \frac{w_2 - w_1}{\pi}$

ODD M SHORTCUT

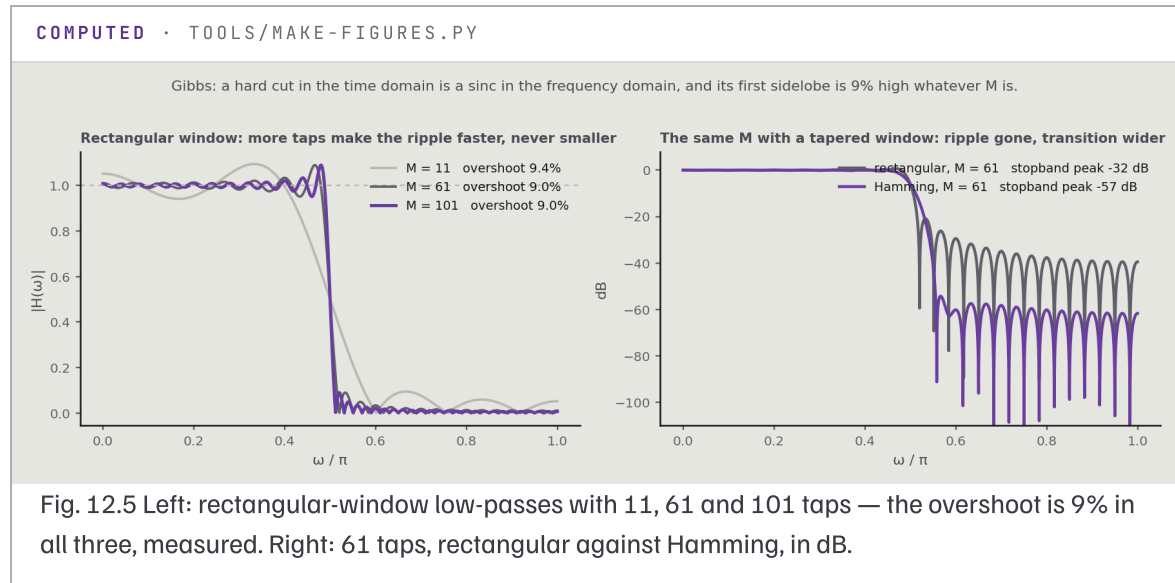
For odd M every $\sin(\pi(n - \tau))$ is zero, so high-pass = $d(n - \tau)$ minus low-pass, and band-stop = $d(n - \tau)$ minus band-pass. Flip every sign, and replace the centre tap by 1 minus itself.

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12.4 Windows, Gibbs, and the table

Cutting the sinc short is multiplying by a box. In frequency that convolves H_d with a sinc whose first side lobe is only 13 dB down. Result: 9% overshoot at every edge, **no matter how big M is** (Gibbs). A window that tapers to zero has lower side lobes and a wider main lobe: less ripple, wider transition. The window sets the stopband floor; M sets the transition width.



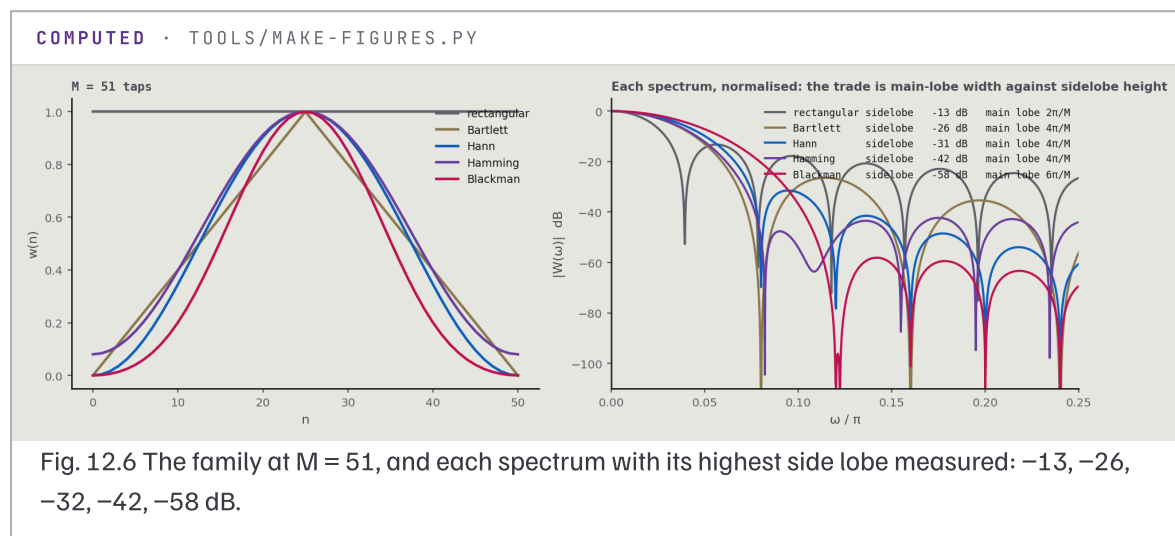
$$\text{Hann: } 0.5 - 0.5 \cos \frac{2\pi n}{M-1}$$

$$\text{Hamming: } 0.54 - 0.46 \cos \frac{2\pi n}{M-1}$$

$$\text{Blackman: } 0.42 - 0.5 \cos \frac{2\pi n}{M-1} + 0.08 \cos \frac{4\pi n}{M-1}$$

$$\text{Bartlett: } 1 - \frac{2|n-\tau|}{M-1}$$

WINDOW	MAIN LOBE	PEAK SIDE LOBE	STOPBAND OF THE FILTER
Rectangular	$\frac{4\pi}{M}$ ($k=2$)	-13 dB	-21 dB
Bartlett	$\frac{8\pi}{M}$ ($k=4$)	-25 dB	-25 dB
Hann	$\frac{8\pi}{M}$ ($k=4$)	-31 dB	-44 dB
Hamming	$\frac{8\pi}{M}$ ($k=4$)	-41 dB	-53 dB
Blackman	$\frac{12\pi}{M}$ ($k=6$)	-57 dB	-74 dB
Kaiser	set by β	set by β	set by β



WORKED EXAMPLE

Low-pass, Hamming window, $\omega_c = \frac{3\pi}{4}$, $M = 7$ (set in June 2015, Dec 2014, June 2012).

$$\tau = 3, \quad h_d(n) = \frac{\sin\left(\frac{3\pi}{4}(n-3)\right)}{\pi(n-3)}, \quad h_d(3) = \frac{\omega_c}{\pi} = \frac{3}{4}$$

N	0	1	2	3	4	5	6
$h_d(n)$	0.075	-0.159	0.225	0.750	0.225	-0.159	0.075
$w(n)$	0.08	0.31	0.77	1.00	0.77	0.31	0.08
$h(n) = h_d(n)w(n)$	0.006	-0.049	0.173	0.750	0.173	-0.049	0.006

$$|H(\omega)| = |0.75 + 0.346 \cos \omega - 0.098 \cos 2\omega + 0.012 \cos 3\omega|$$

Ω	0	0.2π	0.4π	0.5π	0.6π	0.8π	π
$ H $	1.010	0.996	0.927	0.849	0.732	0.443	0.293
dB	0.09	-0.03	-0.66	-1.43	-2.70	-7.08	-10.67

Seven Hamming taps give a transition band about $\frac{8\pi}{7}$ wide — wider than the whole axis — so the filter only reaches -10.7 dB at π . Say that in one line; the question is testing the method. All numbers recomputed with numpy.

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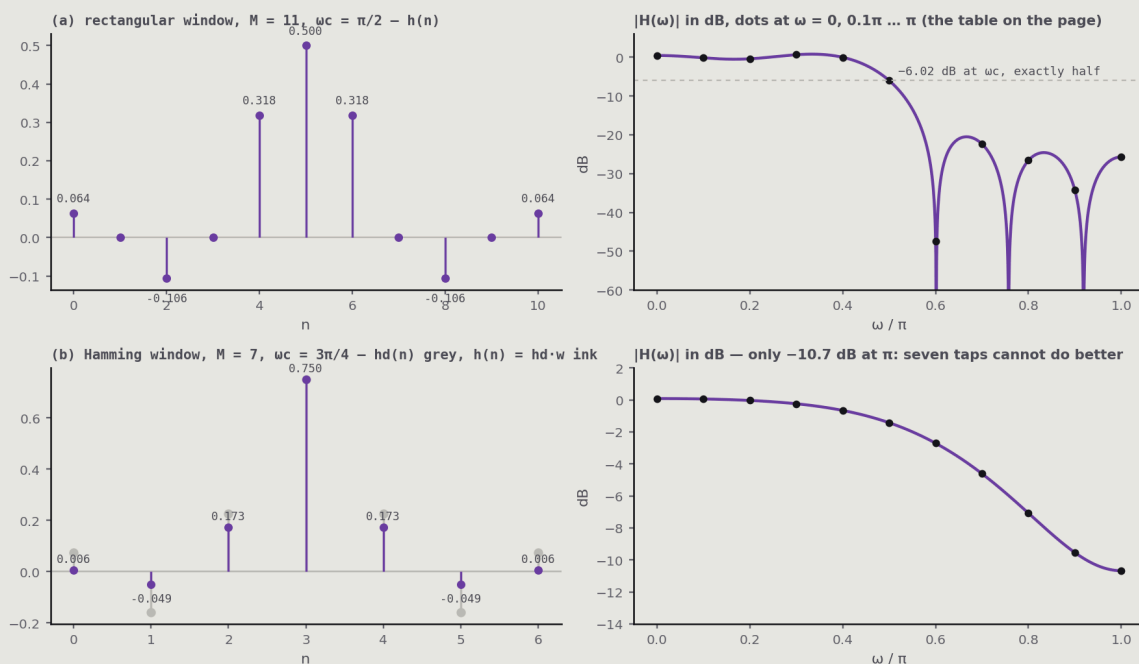


Fig. 12.7 The rectangular $M = 11$ design (top) and this Hamming $M = 7$ design (bottom): coefficients, and $|H|$ in dB with the table frequencies dotted. The top one passes -6.02 dB at ω_c exactly.

THE THREE SLIPS

Forgetting the delay τ in H_d (the answer comes out non-causal). Getting $h_d(\tau)$ wrong — it is $\frac{\omega_c}{\pi}$, by L'Hôpital. And writing M instead of $M-1$ inside the Hamming cosine, which breaks the symmetry $\omega(0) = \omega(M-1)$.

12.5 Choosing M ; the Kaiser recipe

Your notebook's rule: the transition band is about as wide as the window's main lobe, $\frac{k2\pi}{M}$. So $M = \frac{k2\pi}{\omega_s - \omega_p}$ with $k = 2$ for rectangular, 4 for Bartlett, Hann and Hamming, 6 for Blackman. Round up; make it odd for Type 1.

WORKED EXAMPLE

Class notebook: find the order of the filter with passband edge 0.3π and stopband edge 0.45π , rectangular window.

$$\Delta\omega = 0.45\pi - 0.3\pi = 0.15\pi$$

$$M = \frac{k \cdot 2\pi}{\Delta\omega} = \frac{2 \cdot 2\pi}{0.15\pi} = \frac{4}{0.15} = 26.7 \rightarrow M = 27 \text{ taps (order 26)}$$

With a Hamming window ($k = 4$) it would be 54 taps, but the stopband would sit at -53 dB instead of -21 dB.

$$A = -20 \log_{10} \delta, \quad \delta = \min(\delta_p, \delta_s)$$

$$\beta = \begin{cases} 0 & A \leq 21 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21) & 21 < A \leq 50 \\ 0.1102(A - 8.7) & A > 50 \end{cases}$$

$$M \geq \frac{A - 7.95}{14.36 \Delta f}, \quad \Delta f = \frac{\omega_s - \omega_p}{2\pi}, \quad \omega_c = \frac{1}{2}(\omega_p + \omega_s)$$

WORKED EXAMPLE

Kaiser low-pass: $\omega_c = \frac{\pi}{4}$, $dw = 0.02\pi$, $ds = 0.01$.

$$A = -20 \log_{10}(0.01) = 40 \text{ dB}$$

$$\beta = 0.5842(19)^{0.4} + 0.07886(19) = 1.896 + 1.498 = 3.39$$

$$df = \frac{0.02\pi}{2\pi} = 0.01$$

$$M \geq \frac{40 - 7.95}{14.36 \times 0.01} = 223.2 \rightarrow M = 225, \tau = 112$$

$$h(n) = \frac{\frac{\sin(\frac{\pi(n-112)}{4})}{\pi(n-112)} \times I_0\left(3.39\sqrt{1 - \left(\frac{n}{112} - 1\right)^2}\right)}{I_0(3.39)}$$

Two hundred and twenty-five taps for a 1% transition band: that is what a sharp FIR filter costs.

COMPUTED · TOOLS/MAKE-FIGURES.PY

Kaiser: $w(n) = I_0(\beta\sqrt{1 - (2n/(M-1) - 1)^2}) / I_0(\beta)$. Raising β buys stopband attenuation and pays for it with transition width.

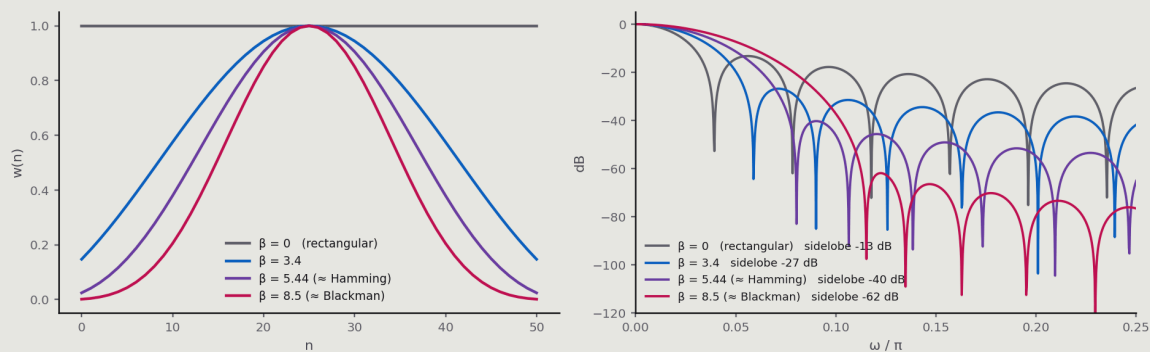


Fig. 12.8 The Kaiser window for four values of β ; 5.44 and 8.5 land on the Hamming and Blackman side-lobe levels.

12.6 Frequency sampling

$$H(k) = H_d\left(\frac{2\pi k}{M}\right), \quad h(n) = \frac{1}{M} \left[H(0) + 2 \sum_{k=1}^P |H(k)| \cos \frac{2\pi k(n-\tau)}{M} \right], \quad P = \frac{M-1}{2} \text{ (odd } M), \frac{M}{2} - 1 \text{ (even)}$$

Take M samples of the ideal response and do the IDFT. The response is **exact at the samples** and ripples between them. For real $h(n)$ the samples must satisfy $H(M - k) = H(k)$; for linear phase give each sample the phase $-\frac{\pi k(M-1)}{M}$.

WORKED EXAMPLE

Low-pass by frequency sampling, $\omega_c = \frac{\pi}{2}$, $M = 17$.

$\tau = 8$, spacing $2\pi/17$, cutoff at $k = 17/4 = 4.25$ $|H(k)| = 1$ for $k = 0..4$, 0 for $k = 5..8$, mirrored for $k = 9..16$ $h(n) = (1/17)[1 + 2 \sum_{k=1}^4 \cos(2\pi k(n-8)/17)]$ $n: 0 \ 1 \ 2 \ 3 \ 4 \ 5 \ 6 \ 7 \ 8$ $h(n): 0.0398 \ -0.0488 \ -0.0346 \ 0.0660 \ 0.0315 \ -0.1075 \ -0.0299 \ 0.3188 \ 0.5294$ check: $h(8) = 9/17 = 0.5294$, and $DFT\{h\} = 1, 1, 1, 1, 1, 0, \dots, 0, 1, 1, 1, 1$ exactly between the samples: passband $+11\%$, stopband peak 0.17 (-15 dB)

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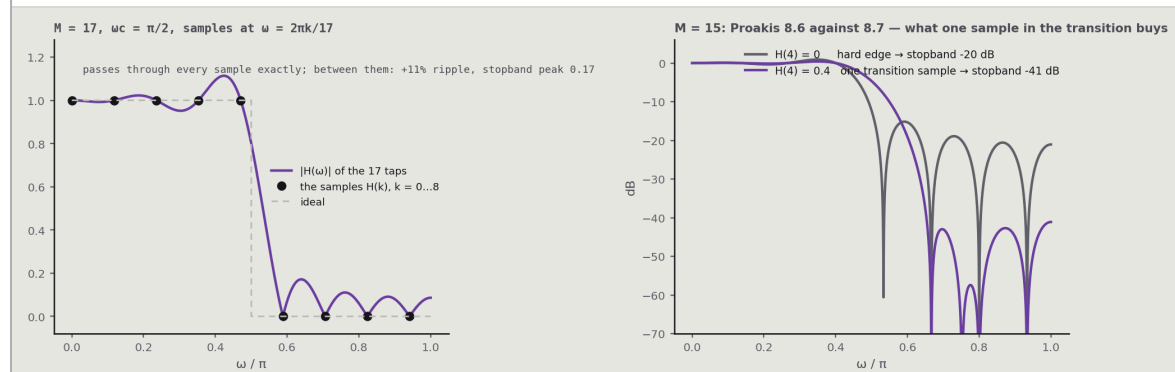


Fig. 12.9 Left: the 17-tap design through its samples, with ripple between. Right: Proakis $M = 15$ with the edge sample 0 (-20 dB) and 0.4 (-41 dB) — one transition sample, twenty decibels.

WINDOW OR FREQUENCY SAMPLING?

Windowing controls the ripple through the window and lets the edges move a little. Frequency sampling fixes M values exactly and lets the ripple move; put one sample in the transition band and the stopband improves by about 20 dB. Equiripple (Parks–McClellan) spreads the error evenly and needs the fewest taps, but it is done by software.

12.7 Differentiator and Hilbert transformer

$$\text{differentiator: } H_d = j\omega, \quad h_d(n) = \frac{\cos \pi(n - \tau)}{n - \tau}; \quad \text{Hilbert: } H_d = -j \operatorname{sgn} \omega, \quad h_d(n) = \frac{1 - \cos \pi(n - \tau)}{\pi(n - \tau)}$$

Both are **antisymmetric** ($h_d(\tau) = 0$), so both are Type 3 or 4 and both have $H(0) = 0$. The differentiator prefers even M (Type 4) so that $H(\pi)$ is not forced to zero; the Hilbert transformer with odd M has every even-offset tap equal to zero.

COMPUTED · TOOLS/MAKE-FIGURES.PY

Both are antisymmetric — $h(n) = -h(M-1-n)$ — so both are Type 3 here, and both have $H(0) = 0$ as they must.

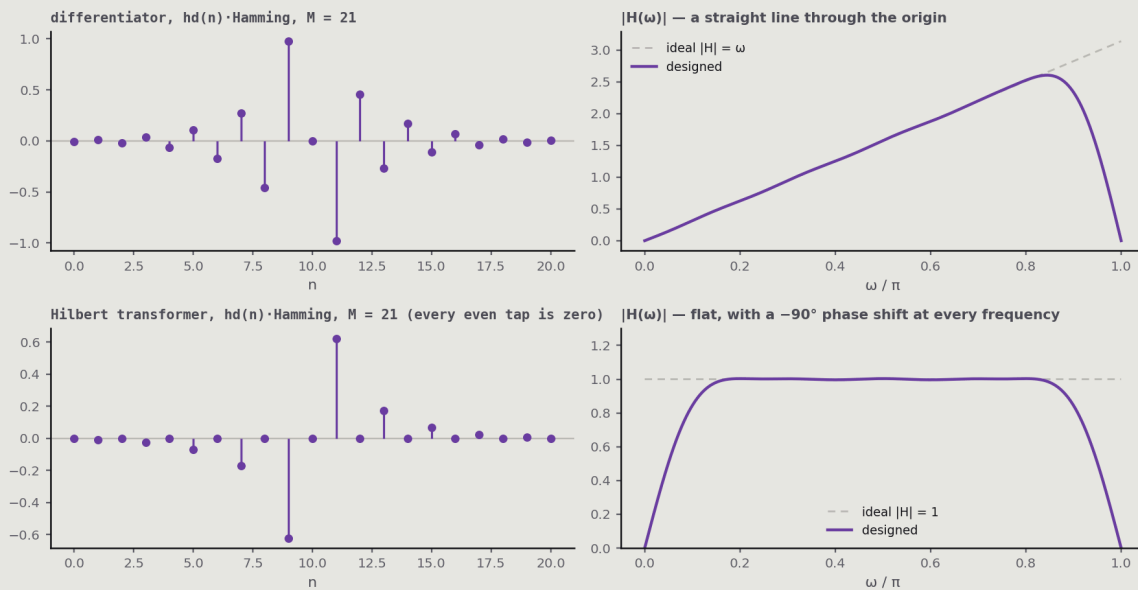


Fig. 12.10 Both at $M = 21$ with a Hamming window: the ramp $|H| = \omega$ and the flat Hilbert magnitude, each rolled off at the ends by the forced zeros.

CHAPTER 12 AT A GLANCE

- FIR: no feedback, always stable, **linear phase exactly when**
 $h(n) = \pm h(M-1-n)$, delay $\frac{M-1}{2}$. Four types; Type 2 has $H(\pi) = 0$, Types 3 and 4 have $H(0) = 0$.
- Window method: $h_d(n) = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$, $h_d(\tau) = \frac{\omega_c}{\pi}$, times $\omega(n)$. HP = $d(n-\tau)$ – LP for odd M .
- Windows: rect –21 dB, Hann –44, Hamming –53, Blackman –74 stopband; main lobe $\frac{k2\pi}{M}$ with $k = 2, 4, 6$. Gibbs = 9% whatever M .
- Length: $M = \frac{k2\pi}{ws-wp}$. Kaiser: $A = -20 \log_{10}(\delta)$, beta from A, $M \geq \frac{A-7.95}{14.36df}$.
- Frequency sampling: $h(n) = \left(\frac{1}{M}\right) \left[H(0) + 2 \sum |H(k)| \cos\left(\frac{2\pi k(n-\tau)}{M}\right) \right]$, exact at the samples; a 0.4 transition sample buys about 20 dB.
- Differentiator $h_d = \frac{\cos(\pi(n-\tau))}{n-\tau}$, Hilbert $h_d = \frac{[1-\cos(\pi(n-\tau))]}{\pi(n-\tau)}$: both antisymmetric, both $H(0) = 0$.

CHAPTER 13

THE PAPER ITSELF

Three mid-sem papers read end to end, and the skeleton does not move: five questions, 30 marks, ninety minutes, and the same four areas every year. This chapter settles what to revise first, what the traps are, and how to spend the ninety minutes.

13.1 What actually gets asked

TOPIC	OF THE 3 PAPERS	WHAT IT LOOKS LIKE
DFT or FFT	3	Compute a small DFT, count operations, or draw a butterfly
Difference equation solved	3	Step response, particular solution, or the full response
z-transform, ROC, causality and stability	3	“Mention the ROC in all cases”
Frequency response	3	Magnitude and phase, or name the filter type
Inverse system	2	Find it, then describe its behaviour
A non-radix-2 FFT	2	Radix-3 flow graph, or split-radix operation counts
Convolution worked by hand	1	Two exponentials, or short lists
Periodicity, energy versus power	1	The four-mark opener
DFT property applied	1	Frequency shift, or symmetry
System identified from its step response	1	Differentiate, transform, divide
Time scaling or shifting inside a z-transform	1	Upsampling combined with an advance
Structure realisation	1	Delay, adder and multiplier blocks
An analog RC circuit analysed as a filter	1	Transfer function, then cut-off

FOUR AREAS CARRY ROUGHLY 24 OF THE 30 MARKS

The **DFT or FFT** question, a **difference equation**, a **z-transform with its ROC**, and a **frequency response**. Every year, without exception. Everything else is the examiner's choice of garnish.

13.2 The one that moves

The FFT question is the unstable one. 2023 asked for a radix-3 flow graph for $N = 9$. 2024 asked for operation counts including split-radix, and separately for DIT applied to the inverse DFT. 2025 asked for decimation in frequency on 16 points.

SO DO NOT LEARN ONLY THE 8-POINT DIT BUTTERFLY

Learn the **counting formulas**, **both decimations**, what changes for the **inverse**, and what changes for a **different radix**. Four small things instead of one big drawing, and they cover every version the question has taken.

13.3 The traps that cost the most marks

MISTAKE	THE CORRECT POSITION
Assuming a discrete sinusoid is always periodic	Only when f is a ratio of whole numbers. $\cos(3n)$ is not periodic.
Losing the arrow — the $n = 0$ position	It fixes where the whole answer sits. Track it through every shift, fold and convolution.
Folding and shifting in the wrong order	$x(-n - k) \neq x(-n + k)$. Rewrite as $x(-(n + k))$ first.
Giving $X(z)$ without the ROC	The ROC is what decides which signal you mean. It is half the marks.
Losing the minus on the anti-causal term	$-a^n u(-n - 1)$, always.
Thinking zeros affect stability	Only poles decide stability.
Treating “give $X(4)$ to $X(7)$ ” as an 8-point DFT	A 4-point DFT has four distinct values and then repeats.
Believing zero padding improves resolution	It only adds dots. Resolution needs a longer record.
Bit-reversing the wrong end of an FFT	DIT takes bit-reversed input ; DIF gives bit-reversed output .
Skipping the free checks	$\sum y = (\sum x)(\sum h)$ for a convolution; $X(0) = \sum x(n)$ and conjugate symmetry for a DFT. Seconds each.

13.4 Ninety minutes, spent

- 1 First five minutes** — read all five questions and start with the DFT or FFT one. It is the most mechanical and the most certain marks in the paper.
- 2 Then the difference equation.** If the question says “using homogeneous and particular solution” you must show both parts — the z-transform alone scores nothing. But do the z-transform on rough paper anyway: two minutes, and it catches the arithmetic slips where these marks are actually lost.
- 3 The z-transform question** — factor, partial-fraction, then spend most of your words on the ROC cases. Write the causal-and-stable sentence out in full.
- 4 The frequency response** — substitute $z = e^{j\omega}$, use $|1 - ae^{-j\omega}|^2 = 1 - 2a \cos \omega + a^2$, then evaluate at $\omega = 0$ and $\omega = \pi$ and name the filter.

- 5 Last ten minutes** — run the free checks. Sum check on every convolution, $X(0) = \sum x(n)$ on every DFT, three samples of the recursion against every closed-form response.

ASSUME AND SAY SO

Several of these questions are genuinely ambiguous without an assumption — which ROC, whether the system is relaxed, whether the sequence starts at $n = 0$. **State the assumption in one line and carry on.** An answer with a stated assumption scores; a blank does not.

CHAPTER 13 AT A GLANCE

- Five questions, 30 marks, ninety minutes, and the same skeleton for three years running.
- Four areas carry about 24 of the 30 marks: **DFT or FFT, a difference equation, a z-transform with its ROC, and a frequency response.**
- The FFT question is the one that moves — learn the counting formulas, both decimations, the inverse, and one non-radix-2 case.
- Most-punished errors: no ROC, the missing minus on the anti-causal term, bit-reversing the wrong end, and treating a repeated DFT value as a longer DFT.
- Spend the last ten minutes on free checks: the convolution sum check, $X(0) = \sum x(n)$, and three samples of the recursion.
- Where a question is ambiguous, state your assumption in one line and answer it. A stated assumption scores; a blank does not.

THE WHOLE SUBJECT IN ONE SCREEN

EVERYTHING, CONDENSED

01 SIGNALS, AND WHAT DISCRETE TIME CHANGES

- Digital frequency $f = \frac{F}{F_s}$; only $-\frac{1}{2} \leq f \leq \frac{1}{2}$ is distinct, and ω and $\omega + 2\pi k$ are the same signal.
- A discrete sinusoid is periodic **only if** f is a ratio of whole numbers; write $f = k/N$ in lowest terms and N is the period.
- For a sum of sinusoids the period is the **LCM** of the individual periods.
- Energy signal $\Rightarrow E$ finite and $P = 0$. Power signal $\Rightarrow E$ infinite and P finite. Every periodic signal is a power signal.
- The fastest discrete oscillation is at $\omega = \pi$: one sign change per sample. There is nothing faster.

02 SAMPLING, ALIASING AND QUANTIZATION

- Sample faster than twice the highest frequency: $F_s > 2F_{\max}$. The Nyquist rate is $2F_{\max}$; the folding frequency is $\frac{F_s}{2}$.
- Every frequency in the family $F_0 + kF_s$ gives identical samples — that is aliasing.
- To find the apparent frequency: form $f = F/F_s$, slide it into $-\frac{1}{2} \leq f \leq \frac{1}{2}$ by whole numbers, multiply back by F_s .
- Aliasing is **not** recoverable: two components can land on the same digital frequency and add.
- Quantization step $D = \frac{x_{\max} - x_{\min}}{L-1}$, error at most $D/2$, and SQNR = 1.76 + 6.02b dB — about **6 dB per bit**.

03 DISCRETE-TIME SIGNALS: WRITING THEM, MOVING THEM

- Four notations for one signal — functional, tabular, sequence, graphical. In sequence form the **arrow marks $n = 0$** and must be carried everywhere.
- $d(n) = u(n) - u(n - 1)$, and $x(n) = \sum_k x(k)d(n - k)$ — any signal is a row of weighted impulses. That identity is where convolution comes from.
- $x(n - k)$ delays to the right for positive k ; $x(-n)$ mirrors about the origin.
- Fold and shift **do not commute**: rewrite as $x(-(n + k))$ first, then fold, then shift by what is left in the bracket.
- Even part $\frac{x(n) + x(-n)}{2}$, odd part $\frac{x(n) - x(-n)}{2}$, and the odd part is always zero at $n = 0$.

04 LTI SYSTEMS AND CONVOLUTION

- For an LTI system, $h(n)$ is everything: **causal** $\Leftrightarrow h(n) = 0$ for $n < 0$, **stable** $\Leftrightarrow \sum |h(n)|$ is finite. FIR is always stable.
- $y(n) = \sum_k x(k)h(n - k)$ — break the input into impulses, add the answers.
- Result length = $n_1 + n_2 - 1$, starting at (start of x + start of h).
- Free check every time: $\sum y = (\sum x)(\sum h)$.
- Cascade multiplies the responses ($h_1 * h_2$); parallel adds them ($h_1 + h_2$).
- Step response is the running sum of h ; so $h(n) = s(n) - s(n - 1)$ recovers the system from its step response.

05 DIFFERENCE EQUATIONS AND THE TOTAL RESPONSE

- No feedback $\rightarrow$ FIR and always stable; feedback $\rightarrow$ IIR and the roots decide stability.
- Total = **natural + forced** = **zero-input + zero-state** — two different cuts of the same answer, and zero-state still contains a transient.
- Homogeneous: solve $\lambda^N + a_1\lambda^{N-1} + \dots = 0$; particular: guess the shape of the input; then fit the constants from $y(0)$, $y(1)$ computed by the equation.
- Impulse response: homogeneous solution with $h_0(0) = 1$, then $h(n) = b_0h_0(n) + b_1h_0(n - 1) + \dots$
- The inverse system swaps x and y — and swaps poles with zeros, so an IIR system inverts to an FIR one.
- Always verify a closed-form answer by running the recursion for three samples. It takes thirty seconds.

06 THE Z-TRANSFORM AND THE ROC

- $X(z) = \sum x(n)z^{-n}$, and the answer is incomplete without its **ROC**.
 - $a^n u(n)$ and $-a^n u(-n-1)$ have the **identical** $X(z)$ and differ only in the ROC — note the minus sign on the left-sided one.
 - ROC is a ring centred on the origin and never contains a pole: right-sided $\rightarrow$ outside the outermost pole, left-sided $\rightarrow$ inside the innermost.
 - Two-sided signals give an annulus, which may be empty — and then no z-transform exists.
 - The workhorse property is the delay: $x(n-k) \leftrightarrow z^{-k} X(z)$, which turns a difference equation into algebra in one line.
 - Convolution becomes multiplication: $x * h \leftrightarrow X(z)H(z)$.
-

07 THE INVERSE Z-TRANSFORM

- Before inverting anything, find the ROC. It decides the sign and the side of every term.
 - Partial fractions: expand $X(z)/z$, split, then multiply back by z — that leaves terms in the $\frac{A}{1-pz^{-1}}$ shape.
 - Term by term: ROC outside the pole $\rightarrow +Ap^n u(n)$; ROC inside the pole $\rightarrow -Ap^n u(-n-1)$, **with the minus sign**.
 - M poles at distinct radii give **$M+1$** possible ROCs and $M+1$ different signals from one $X(z)$.
 - Long division in descending powers of z gives a causal sequence; ascending powers gives an anti-causal one.
 - A complex pole pair $re^{\pm j\omega_0}$ gives $r^n \cos(\omega_0 n + \phi)$ — radius sets the decay, angle sets the oscillation.
-

08 SYSTEM FUNCTION, POLES, ZEROS AND STABILITY

- $H(z)$ = numerator over denominator; **zeros** from the top, **poles** from the bottom, and only poles decide stability.
 - **Causal** $\Leftrightarrow$ ROC is outside the outermost pole. **Stable** $\Leftrightarrow$ ROC contains $|z| = 1$. Two separate tests.
 - **Causal and stable together** $\Leftrightarrow$ every pole strictly inside the unit circle. Nothing else.
 - M distinct pole radii give $M+1$ possible ROCs — a complex conjugate pair is one radius, not two.
 - A pole's radius sets how fast its mode decays; its angle sets the frequency it resonates at.
 - A zero on the unit circle at angle ω_0 puts an exact null in the frequency response at ω_0 .
-

09 FREQUENCY RESPONSE

- Frequency response = $H(z)$ evaluated at $z = e^{j\omega}$, legal only if the ROC contains the unit circle.
- The DTFT is **continuous in ω and periodic with period 2π** — only $-\pi$ to π is distinct.
- Use $|1 - ae^{-j\omega}|^2 = 1 - 2a \cos \omega + a^2$; take magnitudes and angles of numerator and denominator separately.
- Filter type in two substitutions: $H(+1)$ is the DC gain, $H(-1)$ is the gain at $\omega = \pi$. Bigger at $+1$ means low-pass.
- A zero on the unit circle gives an exact null; a pole on it gives infinite gain and no BIBO stability.
- Steady state: a sinusoid in gives the **same frequency** out, scaled by $|H|$ and shifted by $\angle H$.

10 THE DFT AND ITS PROPERTIES

- $X(k) = \sum x(n)W_N^{nk}$ with $W_N = e^{-\frac{j2\pi}{N}}$; for $N = 4$ that is just powers of $-j$.
- Two free checks on every DFT: $X(0) = \sum x(n)$, and for real $x(n)$ $X(N - k) = \text{conj}(X(k))$ with $X(\frac{N}{2})$ purely real.
- The DFT is **periodic in k with period N** — asking for $X(4) \dots X(7)$ of a 4-point DFT is asking for $X(0) \dots X(3)$ again.
- Multiplying two DFTs gives **circular** convolution; the tail wraps round and adds to the front.
- Zero-pad both signals to $L \geq N + M - 1$ and circular convolution equals the linear answer.
- Zero padding adds dots, not resolution. Resolution needs a longer record; bin spacing is $\frac{F_s}{N}$.

11 THE FFT

- The FFT computes the same DFT; the saving comes from $W_N^{k+\frac{N}{2}} = -W_N^k$ and $W_N^{k+N} = W_N^k$.
- Radix-2 costs $(\frac{N}{2}) \log_2 N$ multiplications and $N \log_2 N$ additions, in $\log_2 N$ stages of $N/2$ butterflies — against N^2 and $N(N - 1)$ direct.
- At $N = 8$ that is 12 multiplications instead of 64 (5.3×); at $N = 1024$ it is 5,120 instead of 1,048,576 (205×).
- DIT** splits the input, twiddle before the add, **bit-reversed input**. **DIF** splits the output, twiddle after the add, **bit-reversed output**. Identical cost.
- Both work **in place** in about $5N/2$ words — bit reversal is the price paid for that.
- The IFFT is the same flow graph with conjugated twiddles and a final divide by N ; the operation count is unchanged.

12 FIR FILTER DESIGN

- FIR: no feedback, always stable, **linear phase exactly when** $h(n) = \pm h(M - 1 - n)$, delay $\frac{M-1}{2}$. Four types; Type 2 has $H(\pi) = 0$, Types 3 and 4 have $H(0) = 0$.
- Window method: $h_d(n) = \frac{\sin(\omega_c(n-\tau))}{\pi(n-\tau)}$, $h_d(\tau) = \frac{\omega_c}{\pi}$, times $\omega(n)$. HP = $d(n - \tau)$ – LP for odd M.
- Windows: rect –21 dB, Hann –44, Hamming –53, Blackman –74 stopband; main lobe $\frac{k2\pi}{M}$ with $k = 2, 4, 6$. Gibbs = 9% whatever M.
- Length: $M = \frac{k2\pi}{ws-wp}$. Kaiser: $A = -20 \log_{10}(\delta)$, beta from A, $M \geq \frac{A-7.95}{14.36df}$.
- Frequency sampling: $h(n) = \left(\frac{1}{M}\right) \left[H(0) + 2 \sum |H(k)| \cos\left(\frac{2\pi k(n-\tau)}{M}\right) \right]$, exact at the samples; a 0.4 transition sample buys about 20 dB.
- Differentiator $h_d = \frac{\cos(\pi(n-\tau))}{n-\tau}$, Hilbert $h_d = \frac{[1-\cos(\pi(n-\tau))]}{\pi(n-\tau)}$: both antisymmetric, both $H(0) = 0$.

13 THE PAPER ITSELF

- Five questions, 30 marks, ninety minutes, and the same skeleton for three years running.
- Four areas carry about 24 of the 30 marks: **DFT or FFT**, a **difference equation**, a **z-transform with its ROC**, and a **frequency response**.
- The FFT question is the one that moves — learn the counting formulas, both decimations, the inverse, and one non-radix-2 case.
- Most-punished errors: no ROC, the missing minus on the anti-causal term, bit-reversing the wrong end, and treating a repeated DFT value as a longer DFT.
- Spend the last ten minutes on free checks: the convolution sum check, $X(0) = \sum x(n)$, and three samples of the recursion.
- Where a question is ambiguous, state your assumption in one line and answer it. A stated assumption scores; a blank does not.

That is the entire revision set — 75 lines. If a line does not make sense, the chapter it came from is one tap away, and the **full course** is behind that.